

PATIENT & SPECIMEN INFORMATION

Patient name	Hemingway, Ernest M.	Accession	NGS-2026-052101
Date of birth	1961-07-22 (Age 64)	Specimen	Oral swab, left lateral tongue
MRN	10345621	Collected	2026-05-19
Ordering physician	Dr. Wisdom, DDS MD	Received	2026-05-19
Clinical indication	Oral lesion — left lateral tongue	Reported	2026-05-20

RESULT: VARIANTS DETECTED

DETECTED VARIANTS

Gene	HGVS (coding)	Protein change	VAF	Depth	Tier	Classification
KRAS	NM_004985.5:c.35G>T	p.Gly12Asp	0.43%	2,778x	I	Pathogenic
PIK3CA	NM_006218.4:c.3140A>G	p.His1047Arg	0.70%	4,293x	I	Pathogenic

VARIANT ANNOTATION

KRAS p.Gly12Asp | Pathogenic (Tier I)

KRAS p.Gly12Asp (G12D) is one of the most frequently detected oncogenic driver mutations in human cancer. It locks KRAS in the GTP-bound active state, constitutively activating downstream RAS/MAPK and PI3K/AKT/mTOR proliferative signalling. In oral squamous cell carcinoma, KRAS mutations are found in 3-8% of cases and mark a subgroup with aggressive behaviour. This variant is classified Pathogenic (Tier I, COSMIC ID COSM521). The co-occurrence with PIK3CA H1047R (see below) indicates dual activation of RAS and PI3K pathways, a combination associated with resistance to single-agent targeted therapies and may influence enrolment eligibility in clinical trials.

PIK3CA p.His1047Arg | Pathogenic (Tier I)

PIK3CA p.His1047Arg is a hotspot activating mutation in the kinase domain of the p110α catalytic subunit of PI3-kinase. It constitutively activates the PI3K/AKT/mTOR pathway, promoting cell survival, proliferation, and tumour angiogenesis. PIK3CA mutations occur in 15-20% of OSCC, most commonly at E545K and H1047R hotspots. This variant is classified Pathogenic (Tier I, COSMIC ID COSM775). PIK3CA H1047R is potentially actionable: alpelisib (PI3Kα inhibitor) is FDA-approved for PIK3CA-mutated breast cancer and is under investigation in HNSCC. Multidisciplinary tumour board review for consideration of PI3K-directed therapy or clinical trial enrolment is recommended.

GENES WITH NO PATHOGENIC OR LIKELY PATHOGENIC VARIANTS DETECTED

AJUBA, CASP8, CDKN2A, CUL3, DPYD, EGFR, EP300, FAT1, FBXW7, HRAS, KEAP1, KMT2D, NFE2L2, NOTCH1, NSD1, PIK3R1, PTEN, TERT, TP53
Variants of uncertain significance (Tier III) are not reported.

METHOD SUMMARY

Cell-free and cellular DNA extracted from oral swab is subjected to hybridization capture using the IDT xGen Duplex UMI panel targeting 251 genomic regions (~77 kbp) across 21 cancer-relevant genes. Libraries incorporate 9-bp Unique Molecular Identifiers (UMIs) in a 9M1S+T read structure on both R1 and R2. Libraries are sequenced on Illumina NextSeq (2x150 bp). After alignment to GRCh38 (canonical chromosomes), reads are grouped by UMI using fgbio (adjacency strategy, ≤1 edit distance), and molecular consensus sequences are called requiring ≥2 reads per UMI family. Somatic variants are called on the consensus BAM with VarDict at minimum allele frequency 0.001. Variants ≥0.2% consensus VAF with ≥2 consensus supporting reads are reported. UMI deduplication reduces sequencing noise by >800-fold. Run-level negative controls are included on every sequencing run.

LIMITATIONS

This assay detects somatic single nucleotide variants and small indels within 251 target regions (~77 kbp) at VAF ≥0.2% with validated sensitivity of 98.4% at ≥20,000x raw input depth (~4,000x consensus after UMI deduplication). Structural variants, copy number alterations, gene fusions, large indels, intronic variants outside defined capture regions, and variants in regions with low hybridization capture efficiency are not reliably assessed. A negative result does not exclude malignancy. Tumour heterogeneity, low tumour cell content, and specimen quality may affect variant allele fraction. Variants below the 0.2% VAF reporting threshold may be present but are not reported. At 20,000x raw sequencing depth (~4,000x mean consensus), approximately 0.3% of targets may fall below 1,000x consensus depth; per-target coverage is provided in the laboratory technical report. Results should be interpreted in the context of clinical and pathological findings by a qualified clinician.

Reported by: Dr. Wisdom, MD PhD, FCAP

Report date: 2026-05-20

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

HOTSPOT SCREENING SUMMARY

Patient:

Hemingway, Ernest M.

Accession:

NGS-2026-052101

Hotspots listed below were explicitly interrogated in addition to all variants above 0.2% VAF with ≥2 consensus reads across all 251 targets. Coverage criteria (≥1,000× consensus depth) were met at all targets.

EGFR			Cov	✓
L858R	c.2573T>G	Not detected		
T790M	c.2369C>T	Not detected		
Exon 19 del	c.2235_2249del	Not detected		
KRAS			Cov	✓
G12D	c.35G>T	● DETECTED		
G12V	c.35G>A	Not detected		
G12C	c.34G>T	Not detected		
G13D	c.38G>A	Not detected		
PIK3CA			Cov	✓
E542K	c.1624G>A	Not detected		
E545K	c.1633G>A	Not detected		
H1047R	c.3140A>G	● DETECTED		
TERT			Cov	✓
C228T	c.-124C>T	Not detected		
C250T	c.-146C>T	Not detected		
HRAS			Cov	✓
G12V	c.35G>T	Not detected		
G12D	c.35G>A	Not detected		
Q61L	c.182A>T	Not detected		
Q61R	c.182A>G	Not detected		
NOTCH1			Cov	✓
P2415L	c.7244C>T	Not detected		
R1699W	c.5095C>T	Not detected		
TP53			Cov	✓
R175H	c.524G>A	Not detected		
R248Q	c.743G>A	Not detected		
R248W	c.742C>T	Not detected		
R273H	c.818G>A	Not detected		
R282W	c.844C>T	Not detected		

REMAINING PANEL GENES — SEQUENCE-LEVEL DETECTION, NO PREDEFINED HOTSPOT LIST

AJUBA · CASP8 · CDKN2A · CUL3 · DPYD · EP300 · FAT1 · FBXW7 · KEAP1 · KMT2D · NFE2L2 · NSD1 · PIK3R1 · PTEN

All variants above 0.2% VAF with ≥2 consensus reads are reported for these genes. No variants meeting reporting criteria were detected. All targets in these genes met minimum coverage criteria (≥1,000× consensus depth).

Reported by: Dr. Wisdom, MD PhD, FCAP

Report date: 2026-05-20

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FULL PANEL COVERAGE REPORT — Hemingway, Ernest M. (NGS-2026-052101)

251 targets, 77,417 bp total. Mean consensus depth: 3,667x. Targets below 1,000x consensus: 0 / 251 (0%). Depths are from the UMI-deduplicated consensus BAM (min-reads ≥ 2, 20,000x raw input). Positions reported 1-based.

Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
DPYD				
DPYD	chr1:97,082,291-97,082,491	201	3,381x	✓ Covered
DPYD	chr1:97,449,958-97,450,158	201	3,440x	✓ Covered
DPYD	chr1:97,515,687-97,515,887	201	3,406x	✓ Covered
DPYD	chr1:97,573,763-97,573,963	201	3,387x	✓ Covered
PTEN				
PTEN	chr10:87,864,450-87,864,568	119	2,281x	✓ Covered
PTEN	chr10:87,894,005-87,894,129	125	2,316x	✓ Covered
PTEN	chr10:87,925,493-87,925,577	85	1,708x	✓ Covered
PTEN	chr10:87,931,026-87,931,109	84	1,688x	✓ Covered
PTEN	chr10:87,932,993-87,933,271	279	4,096x	✓ Covered
PTEN	chr10:87,952,098-87,952,279	182	3,198x	✓ Covered
PTEN	chr10:87,957,833-87,958,039	207	3,479x	✓ Covered
PTEN	chr10:87,960,874-87,961,138	265	4,054x	✓ Covered
PTEN	chr10:87,965,267-87,965,489	223	3,473x	✓ Covered
HRAS				
HRAS	chr11:532,619-532,775	157	2,847x	✓ Covered
HRAS	chr11:533,433-533,632	200	3,381x	✓ Covered
HRAS	chr11:533,746-533,964	219	4,273x	✓ Covered
HRAS	chr11:534,192-534,342	151	2,775x	✓ Covered
KRAS				
KRAS	chr12:25,209,778-25,209,931	154	2,803x	✓ Covered
KRAS	chr12:25,225,594-25,225,793	200	3,376x	✓ Covered
KRAS	chr12:25,227,214-25,227,432	219	3,621x	✓ Covered
KRAS	chr12:25,245,254-25,245,404	151	2,778x	● VARIANT DETECTED
KMT2D				
KMT2D	chr12:49,021,763-49,021,892	130	2,406x	✓ Covered
KMT2D	chr12:49,022,023-49,022,171	149	3,294x	✓ Covered
KMT2D	chr12:49,022,260-49,022,373	114	3,752x	✓ Covered
KMT2D	chr12:49,022,570-49,022,895	326	4,376x	✓ Covered
KMT2D	chr12:49,024,558-49,024,728	171	3,054x	✓ Covered
KMT2D	chr12:49,024,790-49,024,966	177	4,660x	✓ Covered
KMT2D	chr12:49,026,162-49,027,342	1,181	5,754x	✓ Covered
KMT2D	chr12:49,027,783-49,027,950	168	2,973x	✓ Covered
KMT2D	chr12:49,027,989-49,028,161	173	5,123x	✓ Covered
KMT2D	chr12:49,028,808-49,028,978	171	3,050x	✓ Covered
KMT2D	chr12:49,029,041-49,029,256	216	4,842x	✓ Covered
KMT2D	chr12:49,029,381-49,029,496	116	3,088x	✓ Covered
KMT2D	chr12:49,030,260-49,030,459	200	3,443x	✓ Covered
KMT2D	chr12:49,030,581-49,030,788	208	4,055x	✓ Covered
KMT2D	chr12:49,030,873-49,031,053	181	4,336x	✓ Covered
KMT2D	chr12:49,031,155-49,033,984	2,830	6,125x	✓ Covered
KMT2D	chr12:49,034,047-49,034,319	273	5,327x	✓ Covered
KMT2D	chr12:49,034,390-49,034,496	107	4,350x	✓ Covered
KMT2D	chr12:49,034,562-49,034,686	125	4,110x	✓ Covered
KMT2D	chr12:49,034,792-49,034,955	164	3,766x	✓ Covered
KMT2D	chr12:49,037,105-49,039,009	1,905	5,939x	✓ Covered
KMT2D	chr12:49,039,202-49,039,378	177	3,211x	✓ Covered
KMT2D	chr12:49,039,415-49,039,637	223	5,323x	✓ Covered

FULL PANEL COVERAGE REPORT — Hemingway, Ernest M. (NGS-2026-052101)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
KMT2D	chr12:49,039,704–49,041,555	1,852	6,094×	✓ Covered
KMT2D	chr12:49,041,635–49,041,725	91	4,502×	✓ Covered
KMT2D	chr12:49,041,897–49,042,010	114	2,367×	✓ Covered
KMT2D	chr12:49,042,069–49,042,350	282	5,059×	✓ Covered
KMT2D	chr12:49,042,541–49,042,665	125	2,457×	✓ Covered
KMT2D	chr12:49,042,721–49,042,898	178	4,739×	✓ Covered
KMT2D	chr12:49,043,056–49,043,206	151	3,045×	✓ Covered
KMT2D	chr12:49,043,343–49,043,448	106	2,786×	✓ Covered
KMT2D	chr12:49,043,615–49,043,802	188	3,413×	✓ Covered
KMT2D	chr12:49,043,848–49,044,018	171	5,012×	✓ Covered
KMT2D	chr12:49,044,180–49,044,324	145	2,863×	✓ Covered
KMT2D	chr12:49,044,383–49,044,542	160	4,617×	✓ Covered
KMT2D	chr12:49,044,724–49,044,985	262	4,062×	✓ Covered
KMT2D	chr12:49,045,900–49,045,987	88	1,704×	✓ Covered
KMT2D	chr12:49,046,045–49,046,194	150	4,280×	✓ Covered
KMT2D	chr12:49,046,240–49,046,444	205	5,105×	✓ Covered
KMT2D	chr12:49,046,589–49,046,810	222	3,937×	✓ Covered
KMT2D	chr12:49,047,945–49,048,089	145	2,646×	✓ Covered
KMT2D	chr12:49,048,639–49,048,789	151	2,707×	✓ Covered
KMT2D	chr12:49,049,085–49,049,238	154	2,742×	✓ Covered
KMT2D	chr12:49,049,662–49,050,810	1,149	5,752×	✓ Covered
KMT2D	chr12:49,050,866–49,052,444	1,579	6,143×	✓ Covered
KMT2D	chr12:49,052,544–49,052,729	186	4,347×	✓ Covered
KMT2D	chr12:49,052,895–49,053,092	198	3,569×	✓ Covered
KMT2D	chr12:49,053,187–49,053,341	155	3,943×	✓ Covered
KMT2D	chr12:49,053,456–49,053,661	206	4,053×	✓ Covered
KMT2D	chr12:49,053,958–49,054,160	203	3,430×	✓ Covered
KMT2D	chr12:49,054,287–49,054,436	150	3,409×	✓ Covered
KMT2D	chr12:49,054,508–49,054,771	264	4,923×	✓ Covered
KMT2D	chr12:49,054,880–49,055,046	167	3,851×	✓ Covered
KMT2D	chr12:49,055,256–49,055,344	89	1,789×	✓ Covered
AJUBA				
AJUBA	chr14:22,973,426–22,973,588	163	2,914×	✓ Covered
AJUBA	chr14:22,974,027–22,974,135	109	2,083×	✓ Covered
AJUBA	chr14:22,974,819–22,974,910	92	1,794×	✓ Covered
AJUBA	chr14:22,974,954–22,975,124	171	4,638×	✓ Covered
AJUBA	chr14:22,976,436–22,976,538	103	1,970×	✓ Covered
AJUBA	chr14:22,976,625–22,976,732	108	3,547×	✓ Covered
AJUBA	chr14:22,978,324–22,978,465	142	2,578×	✓ Covered
AJUBA	chr14:22,981,241–22,982,286	1,046	5,639×	✓ Covered
TP53				
TP53	chr17:7,669,592–7,669,710	119	2,190×	✓ Covered
TP53	chr17:7,670,589–7,670,735	147	2,707×	✓ Covered
TP53	chr17:7,673,515–7,673,628	114	2,145×	✓ Covered
TP53	chr17:7,673,681–7,673,857	177	4,767×	✓ Covered
TP53	chr17:7,674,161–7,674,310	150	2,716×	✓ Covered
TP53	chr17:7,674,839–7,674,991	153	2,742×	✓ Covered
TP53	chr17:7,675,033–7,675,256	224	5,251×	✓ Covered
TP53	chr17:7,675,974–7,676,292	319	4,359×	✓ Covered
TP53	chr17:7,676,362–7,676,423	62	4,280×	✓ Covered
TP53	chr17:7,676,501–7,676,614	114	3,510×	✓ Covered

FULL PANEL COVERAGE REPORT — Hemingway, Ernest M. (NGS-2026-052101)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
KEAP1				
KEAP1	chr19:10,486,635–10,486,838	204	3,429×	✓ Covered
KEAP1	chr19:10,489,172–10,489,388	217	3,629×	✓ Covered
KEAP1	chr19:10,489,628–10,489,873	246	3,809×	✓ Covered
KEAP1	chr19:10,491,557–10,492,282	726	5,387×	✓ Covered
KEAP1	chr19:10,499,375–10,500,053	679	5,320×	✓ Covered
NFE2L2				
NFE2L2	chr2:177,233,985–177,234,291	307	4,315×	✓ Covered
CASP8				
CASP8	chr2:201,266,467–201,266,811	345	4,467×	✓ Covered
CASP8	chr2:201,271,496–201,271,641	146	2,691×	✓ Covered
CASP8	chr2:201,272,618–201,272,796	179	3,133×	✓ Covered
CASP8	chr2:201,272,878–201,272,962	85	3,805×	✓ Covered
CASP8	chr2:201,274,869–201,274,973	105	2,020×	✓ Covered
CASP8	chr2:201,276,807–201,276,988	182	3,164×	✓ Covered
CASP8	chr2:201,284,796–201,285,337	542	5,134×	✓ Covered
CASP8	chr2:201,286,439–201,286,611	173	3,048×	✓ Covered
CUL3				
CUL3	chr2:224,474,228–224,474,396	169	2,999×	✓ Covered
CUL3	chr2:224,478,180–224,478,365	186	3,215×	✓ Covered
CUL3	chr2:224,481,872–224,482,098	227	3,736×	✓ Covered
CUL3	chr2:224,495,812–224,495,986	175	3,105×	✓ Covered
CUL3	chr2:224,497,733–224,497,869	137	2,600×	✓ Covered
CUL3	chr2:224,500,343–224,500,507	165	2,955×	✓ Covered
CUL3	chr2:224,502,945–224,503,092	148	2,653×	✓ Covered
CUL3	chr2:224,503,632–224,503,842	211	3,490×	✓ Covered
CUL3	chr2:224,505,936–224,506,152	217	3,612×	✓ Covered
CUL3	chr2:224,506,838–224,507,023	186	3,282×	✓ Covered
CUL3	chr2:224,511,334–224,511,602	269	4,062×	✓ Covered
CUL3	chr2:224,513,504–224,513,658	155	2,799×	✓ Covered
CUL3	chr2:224,514,592–224,514,792	201	3,457×	✓ Covered
CUL3	chr2:224,535,508–224,535,661	154	2,806×	✓ Covered
CUL3	chr2:224,557,639–224,557,876	238	3,767×	✓ Covered
CUL3	chr2:224,584,924–224,585,029	106	2,026×	✓ Covered
EP300				
EP300	chr22:41,092,985–41,093,118	134	2,456×	✓ Covered
EP300	chr22:41,117,167–41,117,841	675	5,355×	✓ Covered
EP300	chr22:41,125,844–41,126,060	217	3,609×	✓ Covered
EP300	chr22:41,127,467–41,127,768	302	4,248×	✓ Covered
EP300	chr22:41,129,870–41,130,023	154	2,698×	✓ Covered
EP300	chr22:41,131,368–41,131,653	286	4,174×	✓ Covered
EP300	chr22:41,135,793–41,135,926	134	2,462×	✓ Covered
EP300	chr22:41,137,633–41,137,810	178	3,057×	✓ Covered
EP300	chr22:41,140,120–41,140,277	158	2,870×	✓ Covered
EP300	chr22:41,141,028–41,141,242	215	3,576×	✓ Covered
EP300	chr22:41,146,719–41,146,836	118	2,269×	✓ Covered
EP300	chr22:41,147,817–41,147,966	150	2,697×	✓ Covered
EP300	chr22:41,149,018–41,149,195	178	3,092×	✓ Covered
EP300	chr22:41,149,741–41,150,218	478	4,994×	✓ Covered
EP300	chr22:41,151,813–41,152,032	220	3,671×	✓ Covered
EP300	chr22:41,152,186–41,152,370	185	3,449×	✓ Covered

FULL PANEL COVERAGE REPORT — Hemingway, Ernest M. (NGS-2026-052101)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
EP300	chr22:41,154,975–41,155,133	159	2,866×	✓ Covered
EP300	chr22:41,157,149–41,157,428	280	4,100×	✓ Covered
EP300	chr22:41,158,392–41,158,520	129	2,426×	✓ Covered
EP300	chr22:41,160,622–41,160,742	121	2,226×	✓ Covered
EP300	chr22:41,162,703–41,162,799	97	1,848×	✓ Covered
EP300	chr22:41,164,033–41,164,150	118	2,178×	✓ Covered
EP300	chr22:41,166,579–41,166,686	108	2,036×	✓ Covered
EP300	chr22:41,168,429–41,168,619	191	3,303×	✓ Covered
EP300	chr22:41,168,701–41,168,887	187	4,329×	✓ Covered
EP300	chr22:41,169,483–41,169,636	154	2,788×	✓ Covered
EP300	chr22:41,170,386–41,170,591	206	3,462×	✓ Covered
EP300	chr22:41,172,479–41,172,683	205	3,519×	✓ Covered
EP300	chr22:41,173,603–41,173,804	202	3,393×	✓ Covered
EP300	chr22:41,176,227–41,176,548	322	4,381×	✓ Covered
EP300	chr22:41,176,753–41,178,973	2,221	6,007×	✓ Covered
PIK3CA				
PIK3CA	chr3:179,218,190–179,218,354	165	2,923×	✓ Covered
PIK3CA	chr3:179,234,074–179,234,381	308	4,293×	● VARIANT DETECTED
FBXW7				
FBXW7	chr4:152,325,986–152,326,251	266	3,982×	✓ Covered
FAT1				
FAT1	chr4:186,588,575–186,589,240	666	5,312×	✓ Covered
FAT1	chr4:186,595,669–186,595,846	178	3,160×	✓ Covered
FAT1	chr4:186,596,520–186,597,191	672	5,308×	✓ Covered
FAT1	chr4:186,597,662–186,597,812	151	2,692×	✓ Covered
FAT1	chr4:186,597,952–186,598,145	194	3,716×	✓ Covered
FAT1	chr4:186,599,878–186,600,380	503	5,062×	✓ Covered
FAT1	chr4:186,601,249–186,601,446	198	3,329×	✓ Covered
FAT1	chr4:186,602,883–186,603,054	172	3,012×	✓ Covered
FAT1	chr4:186,603,156–186,603,997	842	5,716×	✓ Covered
FAT1	chr4:186,604,357–186,604,594	238	3,816×	✓ Covered
FAT1	chr4:186,606,050–186,606,233	184	3,265×	✓ Covered
FAT1	chr4:186,609,163–186,609,340	178	3,130×	✓ Covered
FAT1	chr4:186,609,781–186,610,035	255	3,973×	✓ Covered
FAT1	chr4:186,611,366–186,611,795	430	4,815×	✓ Covered
FAT1	chr4:186,613,089–186,613,362	274	4,070×	✓ Covered
FAT1	chr4:186,614,171–186,614,364	194	3,336×	✓ Covered
FAT1	chr4:186,616,985–186,617,221	237	3,784×	✓ Covered
FAT1	chr4:186,617,688–186,621,795	4,108	6,164×	✓ Covered
FAT1	chr4:186,628,134–186,628,384	251	3,898×	✓ Covered
FAT1	chr4:186,628,468–186,628,783	316	5,081×	✓ Covered
FAT1	chr4:186,633,664–186,633,843	180	3,154×	✓ Covered
FAT1	chr4:186,636,005–186,636,255	251	3,906×	✓ Covered
FAT1	chr4:186,636,565–186,636,934	370	4,569×	✓ Covered
FAT1	chr4:186,639,702–186,639,803	102	1,967×	✓ Covered
FAT1	chr4:186,663,279–186,663,633	355	4,506×	✓ Covered
FAT1	chr4:186,706,543–186,709,847	3,305	6,128×	✓ Covered
TERT				
TERT	chr5:1,295,050–1,295,250	201	3,422×	✓ Covered
PIK3R1				
PIK3R1	chr5:68,294,516–68,294,698	183	3,230×	✓ Covered

FULL PANEL COVERAGE REPORT — Hemingway, Ernest M. (NGS-2026-052101)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
PIK3R1	chr5:68,295,128-68,295,344	217	3,604×	✓ Covered
PIK3R1	chr5:68,295,400-68,295,508	109	4,680×	✓ Covered
NSD1				
NSD1	chr5:177,135,084-177,136,050	967	5,606×	✓ Covered
NSD1	chr5:177,191,864-177,192,039	176	3,054×	✓ Covered
NSD1	chr5:177,204,100-177,204,312	213	3,526×	✓ Covered
NSD1	chr5:177,209,616-177,212,215	2,600	6,068×	✓ Covered
NSD1	chr5:177,235,801-177,235,965	165	2,957×	✓ Covered
NSD1	chr5:177,238,217-177,238,527	311	4,359×	✓ Covered
NSD1	chr5:177,239,736-177,239,885	150	2,704×	✓ Covered
NSD1	chr5:177,244,175-177,244,290	116	2,204×	✓ Covered
NSD1	chr5:177,246,658-177,246,816	159	2,858×	✓ Covered
NSD1	chr5:177,248,161-177,248,344	184	3,176×	✓ Covered
NSD1	chr5:177,251,710-177,251,873	164	2,904×	✓ Covered
NSD1	chr5:177,256,931-177,257,171	241	3,832×	✓ Covered
NSD1	chr5:177,259,969-177,260,188	220	3,613×	✓ Covered
NSD1	chr5:177,267,542-177,267,738	197	3,354×	✓ Covered
NSD1	chr5:177,269,582-177,269,827	246	3,868×	✓ Covered
NSD1	chr5:177,273,652-177,273,804	153	2,775×	✓ Covered
NSD1	chr5:177,280,545-177,280,854	310	4,324×	✓ Covered
NSD1	chr5:177,282,445-177,282,601	157	2,807×	✓ Covered
NSD1	chr5:177,283,767-177,283,948	182	3,171×	✓ Covered
NSD1	chr5:177,288,799-177,288,945	147	2,689×	✓ Covered
NSD1	chr5:177,291,934-177,292,178	245	3,836×	✓ Covered
NSD1	chr5:177,293,812-177,295,476	1,665	5,891×	✓ Covered
EGFR				
EGFR	chr7:55,173,901-55,174,063	163	2,952×	✓ Covered
EGFR	chr7:55,174,702-55,174,840	139	2,542×	✓ Covered
EGFR	chr7:55,181,273-55,181,498	226	3,670×	✓ Covered
EGFR	chr7:55,191,699-55,191,894	196	3,324×	✓ Covered
CDKN2A				
CDKN2A	chr9:21,968,212-21,968,262	51	1,127×	✓ Covered
CDKN2A	chr9:21,970,882-21,971,228	347	4,489×	✓ Covered
CDKN2A	chr9:21,974,658-21,974,846	189	6,485×	✓ Covered
CDKN2A	chr9:21,974,658-21,974,847	190	6,503×	✓ Covered
NOTCH1				
NOTCH1	chr9:136,496,054-136,497,578	1,525	5,847×	✓ Covered
NOTCH1	chr9:136,498,879-136,499,016	138	2,543×	✓ Covered
NOTCH1	chr9:136,499,092-136,499,279	188	4,412×	✓ Covered
NOTCH1	chr9:136,500,532-136,500,867	336	4,432×	✓ Covered
NOTCH1	chr9:136,501,728-136,501,933	206	3,471×	✓ Covered
NOTCH1	chr9:136,501,981-136,502,108	128	4,923×	✓ Covered
NOTCH1	chr9:136,502,252-136,502,508	257	4,254×	✓ Covered
NOTCH1	chr9:136,503,162-136,503,350	189	3,260×	✓ Covered
NOTCH1	chr9:136,504,653-136,505,124	472	4,906×	✓ Covered
NOTCH1	chr9:136,505,290-136,505,901	612	5,320×	✓ Covered
NOTCH1	chr9:136,506,507-136,506,659	153	2,759×	✓ Covered
NOTCH1	chr9:136,506,696-136,506,993	298	5,481×	✓ Covered
NOTCH1	chr9:136,507,285-136,507,457	173	3,112×	✓ Covered
NOTCH1	chr9:136,507,935-136,508,159	225	3,718×	✓ Covered
NOTCH1	chr9:136,508,212-136,508,405	194	5,001×	✓ Covered

FULL PANEL COVERAGE REPORT — Hemingway, Ernest M. (NGS-2026-052101)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
NOTCH1	chr9:136,508,850-136,509,091	242	3,793x	✓ Covered
NOTCH1	chr9:136,509,713-136,509,981	269	4,037x	✓ Covered
NOTCH1	chr9:136,510,633-136,510,825	193	3,324x	✓ Covered
NOTCH1	chr9:136,511,132-136,511,291	160	2,828x	✓ Covered
NOTCH1	chr9:136,513,001-136,513,154	154	2,743x	✓ Covered
NOTCH1	chr9:136,513,372-136,513,557	186	3,259x	✓ Covered
NOTCH1	chr9:136,514,490-136,514,722	233	3,782x	✓ Covered
NOTCH1	chr9:136,515,270-136,515,420	151	2,743x	✓ Covered
NOTCH1	chr9:136,515,463-136,515,736	274	5,297x	✓ Covered
NOTCH1	chr9:136,515,961-136,516,114	154	2,721x	✓ Covered
NOTCH1	chr9:136,517,252-136,517,405	154	2,798x	✓ Covered
NOTCH1	chr9:136,517,732-136,517,957	226	3,648x	✓ Covered
NOTCH1	chr9:136,518,117-136,518,312	196	3,553x	✓ Covered
NOTCH1	chr9:136,518,571-136,518,844	274	4,077x	✓ Covered
NOTCH1	chr9:136,519,423-136,519,585	163	2,912x	✓ Covered
NOTCH1	chr9:136,522,830-136,523,208	379	4,651x	✓ Covered
NOTCH1	chr9:136,523,697-136,523,999	303	4,324x	✓ Covered
NOTCH1	chr9:136,544,004-136,544,122	119	2,200x	✓ Covered
NOTCH1	chr9:136,545,706-136,545,806	101	1,932x	✓ Covered

PATIENT & SPECIMEN INFORMATION

Patient name	Woolf, Virginia A.	Accession	NGS-2026-052102
Date of birth	1970-03-11 (Age 56)	Specimen	Oral swab, right buccal mucosa
MRN	20456732	Collected	2026-05-19
Ordering physician	Dr. Wisdom, MD	Received	2026-05-19
Clinical indication	Oral lesion — right buccal mucosa	Reported	2026-05-20

RESULT: VARIANTS DETECTED

DETECTED VARIANTS

Gene	HGVS (coding)	Protein change	VAF	Depth	Tier	Classification
NOTCH1	NM_017617.5:c.7244C>T	p.Pro2415Leu	0.26%	5,855x	II	Likely Pathogenic

VARIANT ANNOTATION

NOTCH1 p.Pro2415Leu | Likely Pathogenic (Tier II)

NOTCH1 functions as a tumour suppressor in stratified squamous epithelium. The p.Pro2415Leu missense variant affects a conserved residue in the PEST domain, predicted to disrupt protein stability (SIFT: deleterious; PolyPhen-2: probably damaging). NOTCH1 loss-of-function mutations are found in 10–15% of OSCC and are associated with early-onset tumourigenesis in the oral cavity. This variant is classified Likely Pathogenic (Tier II). In the context of confirmed dysplasia, detection of a NOTCH1 tumour suppressor variant supports malignant progression risk. Clinical correlation and multidisciplinary review are recommended.

GENES WITH NO PATHOGENIC OR LIKELY PATHOGENIC VARIANTS DETECTED

AJUBA, CASP8, CDKN2A, CUL3, DPYD, EGFR, EP300, FAT1, FBXW7, HRAS, KEAP1, KMT2D, KRAS, NFE2L2, NSD1, PIK3CA, PIK3R1, PTEN, TERT, TP53

Variants of uncertain significance (Tier III) are not reported.

METHOD SUMMARY

Cell-free and cellular DNA extracted from oral swab is subjected to hybridization capture using the IDT xGen Duplex UMI panel targeting 251 genomic regions (~77 kbp) across 21 cancer-relevant genes. Libraries incorporate 9-bp Unique Molecular Identifiers (UMIs) in a 9M1S+T read structure on both R1 and R2. Libraries are sequenced on Illumina NextSeq (2x150 bp). After alignment to GRCh38 (canonical chromosomes), reads are grouped by UMI using fgbio (adjacency strategy, ≤1 edit distance), and molecular consensus sequences are called requiring ≥2 reads per UMI family. Somatic variants are called on the consensus BAM with VarDict at minimum allele frequency 0.001. Variants ≥0.2% consensus VAF with ≥2 consensus supporting reads are reported. UMI deduplication reduces sequencing noise by >800-fold. Run-level negative controls are included on every sequencing run.

LIMITATIONS

This assay detects somatic single nucleotide variants and small indels within 251 target regions (~77 kbp) at VAF ≥0.2% with validated sensitivity of 98.4% at ≥20,000x raw input depth (~4,000x consensus after UMI deduplication). Structural variants, copy number alterations, gene fusions, large indels, intronic variants outside defined capture regions, and variants in regions with low hybridization capture efficiency are not reliably assessed. A negative result does not exclude malignancy. Tumour heterogeneity, low tumour cell content, and specimen quality may affect variant allele fraction. Variants below the 0.2% VAF reporting threshold may be present but are not reported. At 20,000x raw sequencing depth (~4,000x mean consensus), approximately 0.3% of targets may fall below 1,000x consensus depth; per-target coverage is provided in the laboratory technical report. Results should be interpreted in the context of clinical and pathological findings by a qualified clinician.

Reported by: Dr. Wisdom, MD PhD, FCAP

Report date: 2026-05-20

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

HOTSPOT SCREENING SUMMARY

Patient:Woolf, Virginia A.

Accession:NGS-2026-052102

Hotspots listed below were explicitly interrogated in addition to all variants above 0.2% VAF with ≥2 consensus reads across all 251 targets. Coverage criteria (≥1,000× consensus depth) were met at all targets.

EGFRCov ✓			HRASCov ✓		
L858R	c.2573T>G	Not detected	G12V	c.35G>T	Not detected
T790M	c.2369C>T	Not detected	G12D	c.35G>A	Not detected
Exon 19 del	c.2235_2249del	Not detected	Q61L	c.182A>T	Not detected
			Q61R	c.182A>G	Not detected
KRASCov ✓			NOTCH1Cov ✓		
G12D	c.35G>T	Not detected	P2415L	c.7244C>T	● DETECTED
G12V	c.35G>A	Not detected	R1699W	c.5095C>T	Not detected
G12C	c.34G>T	Not detected			
G13D	c.38G>A	Not detected	TP53Cov ✓		
PIK3CACov ✓			R175H	c.524G>A	Not detected
E542K	c.1624G>A	Not detected	R248Q	c.743G>A	Not detected
E545K	c.1633G>A	Not detected	R248W	c.742C>T	Not detected
H1047R	c.3140A>G	Not detected	R273H	c.818G>A	Not detected
TERTCov ✓			R282W	c.844C>T	Not detected
C228T	c.-124C>T	Not detected			
C250T	c.-146C>T	Not detected			

REMAINING PANEL GENES — SEQUENCE-LEVEL DETECTION, NO PREDEFINED HOTSPOT LIST

AJUBA · CASP8 · CDKN2A · CUL3 · DPYD · EP300 · FAT1 · FBXW7 · KEAP1 · KMT2D · NFE2L2 · NSD1 · PIK3R1 · PTEN

All variants above 0.2% VAF with ≥2 consensus reads are reported for these genes. No variants meeting reporting criteria were detected. All targets in these genes met minimum coverage criteria (≥1,000× consensus depth).

Reported by: Dr. Wisdom, MD PhD, FCAP

Report date: 2026-05-20

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

FULL PANEL COVERAGE REPORT — Woolf, Virginia A. (NGS-2026-052102)

251 targets, 77,417 bp total. Mean consensus depth: 3,665x. Targets below 1,000x consensus: 0 / 251 (0%). Depths are from the UMI-deduplicated consensus BAM (min-reads ≥ 2, 20,000x raw input). Positions reported 1-based.

Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
DPYD				
DPYD	chr1:97,082,291-97,082,491	201	3,447x	✓ Covered
DPYD	chr1:97,449,958-97,450,158	201	3,492x	✓ Covered
DPYD	chr1:97,515,687-97,515,887	201	3,452x	✓ Covered
DPYD	chr1:97,573,763-97,573,963	201	3,469x	✓ Covered
PTEN				
PTEN	chr10:87,864,450-87,864,568	119	2,266x	✓ Covered
PTEN	chr10:87,894,005-87,894,129	125	2,315x	✓ Covered
PTEN	chr10:87,925,493-87,925,577	85	1,735x	✓ Covered
PTEN	chr10:87,931,026-87,931,109	84	1,637x	✓ Covered
PTEN	chr10:87,932,993-87,933,271	279	4,109x	✓ Covered
PTEN	chr10:87,952,098-87,952,279	182	3,186x	✓ Covered
PTEN	chr10:87,957,833-87,958,039	207	3,517x	✓ Covered
PTEN	chr10:87,960,874-87,961,138	265	4,049x	✓ Covered
PTEN	chr10:87,965,267-87,965,489	223	3,488x	✓ Covered
HRAS				
HRAS	chr11:532,619-532,775	157	2,809x	✓ Covered
HRAS	chr11:533,433-533,632	200	3,417x	✓ Covered
HRAS	chr11:533,746-533,964	219	4,200x	✓ Covered
HRAS	chr11:534,192-534,342	151	2,709x	✓ Covered
KRAS				
KRAS	chr12:25,209,778-25,209,931	154	2,740x	✓ Covered
KRAS	chr12:25,225,594-25,225,793	200	3,379x	✓ Covered
KRAS	chr12:25,227,214-25,227,432	219	3,583x	✓ Covered
KRAS	chr12:25,245,254-25,245,404	151	2,699x	✓ Covered
KMT2D				
KMT2D	chr12:49,021,763-49,021,892	130	2,432x	✓ Covered
KMT2D	chr12:49,022,023-49,022,171	149	3,251x	✓ Covered
KMT2D	chr12:49,022,260-49,022,373	114	3,798x	✓ Covered
KMT2D	chr12:49,022,570-49,022,895	326	4,424x	✓ Covered
KMT2D	chr12:49,024,558-49,024,728	171	3,055x	✓ Covered
KMT2D	chr12:49,024,790-49,024,966	177	4,689x	✓ Covered
KMT2D	chr12:49,026,162-49,027,342	1,181	5,736x	✓ Covered
KMT2D	chr12:49,027,783-49,027,950	168	2,979x	✓ Covered
KMT2D	chr12:49,027,989-49,028,161	173	5,128x	✓ Covered
KMT2D	chr12:49,028,808-49,028,978	171	3,017x	✓ Covered
KMT2D	chr12:49,029,041-49,029,256	216	4,894x	✓ Covered
KMT2D	chr12:49,029,381-49,029,496	116	3,144x	✓ Covered
KMT2D	chr12:49,030,260-49,030,459	200	3,398x	✓ Covered
KMT2D	chr12:49,030,581-49,030,788	208	4,040x	✓ Covered
KMT2D	chr12:49,030,873-49,031,053	181	4,349x	✓ Covered
KMT2D	chr12:49,031,155-49,033,984	2,830	6,140x	✓ Covered
KMT2D	chr12:49,034,047-49,034,319	273	5,350x	✓ Covered
KMT2D	chr12:49,034,390-49,034,496	107	4,455x	✓ Covered
KMT2D	chr12:49,034,562-49,034,686	125	4,131x	✓ Covered
KMT2D	chr12:49,034,792-49,034,955	164	3,734x	✓ Covered
KMT2D	chr12:49,037,105-49,039,009	1,905	5,978x	✓ Covered
KMT2D	chr12:49,039,202-49,039,378	177	3,187x	✓ Covered
KMT2D	chr12:49,039,415-49,039,637	223	5,413x	✓ Covered

FULL PANEL COVERAGE REPORT — Woolf, Virginia A. (NGS-2026-052102)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
KMT2D	chr12:49,039,704–49,041,555	1,852	6,108×	✓ Covered
KMT2D	chr12:49,041,635–49,041,725	91	4,392×	✓ Covered
KMT2D	chr12:49,041,897–49,042,010	114	2,328×	✓ Covered
KMT2D	chr12:49,042,069–49,042,350	282	5,076×	✓ Covered
KMT2D	chr12:49,042,541–49,042,665	125	2,378×	✓ Covered
KMT2D	chr12:49,042,721–49,042,898	178	4,621×	✓ Covered
KMT2D	chr12:49,043,056–49,043,206	151	3,074×	✓ Covered
KMT2D	chr12:49,043,343–49,043,448	106	2,727×	✓ Covered
KMT2D	chr12:49,043,615–49,043,802	188	3,399×	✓ Covered
KMT2D	chr12:49,043,848–49,044,018	171	5,094×	✓ Covered
KMT2D	chr12:49,044,180–49,044,324	145	2,914×	✓ Covered
KMT2D	chr12:49,044,383–49,044,542	160	4,601×	✓ Covered
KMT2D	chr12:49,044,724–49,044,985	262	4,031×	✓ Covered
KMT2D	chr12:49,045,900–49,045,987	88	1,673×	✓ Covered
KMT2D	chr12:49,046,045–49,046,194	150	4,206×	✓ Covered
KMT2D	chr12:49,046,240–49,046,444	205	5,053×	✓ Covered
KMT2D	chr12:49,046,589–49,046,810	222	3,905×	✓ Covered
KMT2D	chr12:49,047,945–49,048,089	145	2,582×	✓ Covered
KMT2D	chr12:49,048,639–49,048,789	151	2,714×	✓ Covered
KMT2D	chr12:49,049,085–49,049,238	154	2,770×	✓ Covered
KMT2D	chr12:49,049,662–49,050,810	1,149	5,674×	✓ Covered
KMT2D	chr12:49,050,866–49,052,444	1,579	6,129×	✓ Covered
KMT2D	chr12:49,052,544–49,052,729	186	4,423×	✓ Covered
KMT2D	chr12:49,052,895–49,053,092	198	3,575×	✓ Covered
KMT2D	chr12:49,053,187–49,053,341	155	3,994×	✓ Covered
KMT2D	chr12:49,053,456–49,053,661	206	4,052×	✓ Covered
KMT2D	chr12:49,053,958–49,054,160	203	3,441×	✓ Covered
KMT2D	chr12:49,054,287–49,054,436	150	3,407×	✓ Covered
KMT2D	chr12:49,054,508–49,054,771	264	4,969×	✓ Covered
KMT2D	chr12:49,054,880–49,055,046	167	3,825×	✓ Covered
KMT2D	chr12:49,055,256–49,055,344	89	1,797×	✓ Covered
AJUBA				
AJUBA	chr14:22,973,426–22,973,588	163	2,853×	✓ Covered
AJUBA	chr14:22,974,027–22,974,135	109	2,052×	✓ Covered
AJUBA	chr14:22,974,819–22,974,910	92	1,796×	✓ Covered
AJUBA	chr14:22,974,954–22,975,124	171	4,627×	✓ Covered
AJUBA	chr14:22,976,436–22,976,538	103	1,907×	✓ Covered
AJUBA	chr14:22,976,625–22,976,732	108	3,525×	✓ Covered
AJUBA	chr14:22,978,324–22,978,465	142	2,602×	✓ Covered
AJUBA	chr14:22,981,241–22,982,286	1,046	5,685×	✓ Covered
TP53				
TP53	chr17:7,669,592–7,669,710	119	2,256×	✓ Covered
TP53	chr17:7,670,589–7,670,735	147	2,675×	✓ Covered
TP53	chr17:7,673,515–7,673,628	114	2,162×	✓ Covered
TP53	chr17:7,673,681–7,673,857	177	4,731×	✓ Covered
TP53	chr17:7,674,161–7,674,310	150	2,768×	✓ Covered
TP53	chr17:7,674,839–7,674,991	153	2,787×	✓ Covered
TP53	chr17:7,675,033–7,675,256	224	5,175×	✓ Covered
TP53	chr17:7,675,974–7,676,292	319	4,362×	✓ Covered
TP53	chr17:7,676,362–7,676,423	62	4,264×	✓ Covered
TP53	chr17:7,676,501–7,676,614	114	3,454×	✓ Covered

FULL PANEL COVERAGE REPORT — Woolf, Virginia A. (NGS-2026-052102)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
KEAP1				
KEAP1	chr19:10,486,635–10,486,838	204	3,492×	✓ Covered
KEAP1	chr19:10,489,172–10,489,388	217	3,618×	✓ Covered
KEAP1	chr19:10,489,628–10,489,873	246	3,821×	✓ Covered
KEAP1	chr19:10,491,557–10,492,282	726	5,412×	✓ Covered
KEAP1	chr19:10,499,375–10,500,053	679	5,348×	✓ Covered
NFE2L2				
NFE2L2	chr2:177,233,985–177,234,291	307	4,321×	✓ Covered
CASP8				
CASP8	chr2:201,266,467–201,266,811	345	4,519×	✓ Covered
CASP8	chr2:201,271,496–201,271,641	146	2,672×	✓ Covered
CASP8	chr2:201,272,618–201,272,796	179	3,129×	✓ Covered
CASP8	chr2:201,272,878–201,272,962	85	3,868×	✓ Covered
CASP8	chr2:201,274,869–201,274,973	105	1,970×	✓ Covered
CASP8	chr2:201,276,807–201,276,988	182	3,160×	✓ Covered
CASP8	chr2:201,284,796–201,285,337	542	5,070×	✓ Covered
CASP8	chr2:201,286,439–201,286,611	173	3,024×	✓ Covered
CUL3				
CUL3	chr2:224,474,228–224,474,396	169	3,019×	✓ Covered
CUL3	chr2:224,478,180–224,478,365	186	3,209×	✓ Covered
CUL3	chr2:224,481,872–224,482,098	227	3,685×	✓ Covered
CUL3	chr2:224,495,812–224,495,986	175	3,148×	✓ Covered
CUL3	chr2:224,497,733–224,497,869	137	2,526×	✓ Covered
CUL3	chr2:224,500,343–224,500,507	165	2,905×	✓ Covered
CUL3	chr2:224,502,945–224,503,092	148	2,656×	✓ Covered
CUL3	chr2:224,503,632–224,503,842	211	3,515×	✓ Covered
CUL3	chr2:224,505,936–224,506,152	217	3,660×	✓ Covered
CUL3	chr2:224,506,838–224,507,023	186	3,208×	✓ Covered
CUL3	chr2:224,511,334–224,511,602	269	4,074×	✓ Covered
CUL3	chr2:224,513,504–224,513,658	155	2,779×	✓ Covered
CUL3	chr2:224,514,592–224,514,792	201	3,422×	✓ Covered
CUL3	chr2:224,535,508–224,535,661	154	2,740×	✓ Covered
CUL3	chr2:224,557,639–224,557,876	238	3,742×	✓ Covered
CUL3	chr2:224,584,924–224,585,029	106	2,075×	✓ Covered
EP300				
EP300	chr22:41,092,985–41,093,118	134	2,472×	✓ Covered
EP300	chr22:41,117,167–41,117,841	675	5,340×	✓ Covered
EP300	chr22:41,125,844–41,126,060	217	3,561×	✓ Covered
EP300	chr22:41,127,467–41,127,768	302	4,193×	✓ Covered
EP300	chr22:41,129,870–41,130,023	154	2,741×	✓ Covered
EP300	chr22:41,131,368–41,131,653	286	4,139×	✓ Covered
EP300	chr22:41,135,793–41,135,926	134	2,478×	✓ Covered
EP300	chr22:41,137,633–41,137,810	178	3,079×	✓ Covered
EP300	chr22:41,140,120–41,140,277	158	2,831×	✓ Covered
EP300	chr22:41,141,028–41,141,242	215	3,602×	✓ Covered
EP300	chr22:41,146,719–41,146,836	118	2,254×	✓ Covered
EP300	chr22:41,147,817–41,147,966	150	2,752×	✓ Covered
EP300	chr22:41,149,018–41,149,195	178	3,089×	✓ Covered
EP300	chr22:41,149,741–41,150,218	478	4,910×	✓ Covered
EP300	chr22:41,151,813–41,152,032	220	3,637×	✓ Covered
EP300	chr22:41,152,186–41,152,370	185	3,488×	✓ Covered

FULL PANEL COVERAGE REPORT — Woolf, Virginia A. (NGS-2026-052102)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
EP300	chr22:41,154,975-41,155,133	159	2,864×	✓ Covered
EP300	chr22:41,157,149-41,157,428	280	4,101×	✓ Covered
EP300	chr22:41,158,392-41,158,520	129	2,380×	✓ Covered
EP300	chr22:41,160,622-41,160,742	121	2,254×	✓ Covered
EP300	chr22:41,162,703-41,162,799	97	1,911×	✓ Covered
EP300	chr22:41,164,033-41,164,150	118	2,197×	✓ Covered
EP300	chr22:41,166,579-41,166,686	108	2,060×	✓ Covered
EP300	chr22:41,168,429-41,168,619	191	3,321×	✓ Covered
EP300	chr22:41,168,701-41,168,887	187	4,397×	✓ Covered
EP300	chr22:41,169,483-41,169,636	154	2,788×	✓ Covered
EP300	chr22:41,170,386-41,170,591	206	3,482×	✓ Covered
EP300	chr22:41,172,479-41,172,683	205	3,530×	✓ Covered
EP300	chr22:41,173,603-41,173,804	202	3,419×	✓ Covered
EP300	chr22:41,176,227-41,176,548	322	4,352×	✓ Covered
EP300	chr22:41,176,753-41,178,973	2,221	5,999×	✓ Covered
PIK3CA				
PIK3CA	chr3:179,218,190-179,218,354	165	2,906×	✓ Covered
PIK3CA	chr3:179,234,074-179,234,381	308	4,335×	✓ Covered
FBXW7				
FBXW7	chr4:152,325,986-152,326,251	266	3,985×	✓ Covered
FAT1				
FAT1	chr4:186,588,575-186,589,240	666	5,332×	✓ Covered
FAT1	chr4:186,595,669-186,595,846	178	3,147×	✓ Covered
FAT1	chr4:186,596,520-186,597,191	672	5,349×	✓ Covered
FAT1	chr4:186,597,662-186,597,812	151	2,700×	✓ Covered
FAT1	chr4:186,597,952-186,598,145	194	3,665×	✓ Covered
FAT1	chr4:186,599,878-186,600,380	503	5,026×	✓ Covered
FAT1	chr4:186,601,249-186,601,446	198	3,399×	✓ Covered
FAT1	chr4:186,602,883-186,603,054	172	3,057×	✓ Covered
FAT1	chr4:186,603,156-186,603,997	842	5,691×	✓ Covered
FAT1	chr4:186,604,357-186,604,594	238	3,833×	✓ Covered
FAT1	chr4:186,606,050-186,606,233	184	3,186×	✓ Covered
FAT1	chr4:186,609,163-186,609,340	178	3,106×	✓ Covered
FAT1	chr4:186,609,781-186,610,035	255	3,909×	✓ Covered
FAT1	chr4:186,611,366-186,611,795	430	4,793×	✓ Covered
FAT1	chr4:186,613,089-186,613,362	274	4,074×	✓ Covered
FAT1	chr4:186,614,171-186,614,364	194	3,334×	✓ Covered
FAT1	chr4:186,616,985-186,617,221	237	3,711×	✓ Covered
FAT1	chr4:186,617,688-186,621,795	4,108	6,168×	✓ Covered
FAT1	chr4:186,628,134-186,628,384	251	3,895×	✓ Covered
FAT1	chr4:186,628,468-186,628,783	316	5,071×	✓ Covered
FAT1	chr4:186,633,664-186,633,843	180	3,147×	✓ Covered
FAT1	chr4:186,636,005-186,636,255	251	3,882×	✓ Covered
FAT1	chr4:186,636,565-186,636,934	370	4,631×	✓ Covered
FAT1	chr4:186,639,702-186,639,803	102	1,970×	✓ Covered
FAT1	chr4:186,663,279-186,663,633	355	4,567×	✓ Covered
FAT1	chr4:186,706,543-186,709,847	3,305	6,115×	✓ Covered
TERT				
TERT	chr5:1,295,050-1,295,250	201	3,405×	✓ Covered
PIK3R1				
PIK3R1	chr5:68,294,516-68,294,698	183	3,173×	✓ Covered

FULL PANEL COVERAGE REPORT — Woolf, Virginia A. (NGS-2026-052102)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
PIK3R1	chr5:68,295,128-68,295,344	217	3,678x	✓ Covered
PIK3R1	chr5:68,295,400-68,295,508	109	4,568x	✓ Covered
NSD1				
NSD1	chr5:177,135,084-177,136,050	967	5,610x	✓ Covered
NSD1	chr5:177,191,864-177,192,039	176	3,087x	✓ Covered
NSD1	chr5:177,204,100-177,204,312	213	3,577x	✓ Covered
NSD1	chr5:177,209,616-177,212,215	2,600	6,062x	✓ Covered
NSD1	chr5:177,235,801-177,235,965	165	2,957x	✓ Covered
NSD1	chr5:177,238,217-177,238,527	311	4,266x	✓ Covered
NSD1	chr5:177,239,736-177,239,885	150	2,656x	✓ Covered
NSD1	chr5:177,244,175-177,244,290	116	2,151x	✓ Covered
NSD1	chr5:177,246,658-177,246,816	159	2,791x	✓ Covered
NSD1	chr5:177,248,161-177,248,344	184	3,223x	✓ Covered
NSD1	chr5:177,251,710-177,251,873	164	2,915x	✓ Covered
NSD1	chr5:177,256,931-177,257,171	241	3,852x	✓ Covered
NSD1	chr5:177,259,969-177,260,188	220	3,622x	✓ Covered
NSD1	chr5:177,267,542-177,267,738	197	3,379x	✓ Covered
NSD1	chr5:177,269,582-177,269,827	246	3,834x	✓ Covered
NSD1	chr5:177,273,652-177,273,804	153	2,695x	✓ Covered
NSD1	chr5:177,280,545-177,280,854	310	4,311x	✓ Covered
NSD1	chr5:177,282,445-177,282,601	157	2,852x	✓ Covered
NSD1	chr5:177,283,767-177,283,948	182	3,196x	✓ Covered
NSD1	chr5:177,288,799-177,288,945	147	2,636x	✓ Covered
NSD1	chr5:177,291,934-177,292,178	245	3,907x	✓ Covered
NSD1	chr5:177,293,812-177,295,476	1,665	5,905x	✓ Covered
EGFR				
EGFR	chr7:55,173,901-55,174,063	163	2,949x	✓ Covered
EGFR	chr7:55,174,702-55,174,840	139	2,511x	✓ Covered
EGFR	chr7:55,181,273-55,181,498	226	3,710x	✓ Covered
EGFR	chr7:55,191,699-55,191,894	196	3,333x	✓ Covered
CDKN2A				
CDKN2A	chr9:21,968,212-21,968,262	51	1,105x	✓ Covered
CDKN2A	chr9:21,970,882-21,971,228	347	4,479x	✓ Covered
CDKN2A	chr9:21,974,658-21,974,846	189	6,606x	✓ Covered
CDKN2A	chr9:21,974,658-21,974,847	190	6,624x	✓ Covered
NOTCH1				
NOTCH1	chr9:136,496,054-136,497,578	1,525	5,855x	● VARIANT DETECTED
NOTCH1	chr9:136,498,879-136,499,016	138	2,546x	✓ Covered
NOTCH1	chr9:136,499,092-136,499,279	188	4,411x	✓ Covered
NOTCH1	chr9:136,500,532-136,500,867	336	4,444x	✓ Covered
NOTCH1	chr9:136,501,728-136,501,933	206	3,399x	✓ Covered
NOTCH1	chr9:136,501,981-136,502,108	128	4,868x	✓ Covered
NOTCH1	chr9:136,502,252-136,502,508	257	4,253x	✓ Covered
NOTCH1	chr9:136,503,162-136,503,350	189	3,266x	✓ Covered
NOTCH1	chr9:136,504,653-136,505,124	472	4,940x	✓ Covered
NOTCH1	chr9:136,505,290-136,505,901	612	5,252x	✓ Covered
NOTCH1	chr9:136,506,507-136,506,659	153	2,747x	✓ Covered
NOTCH1	chr9:136,506,696-136,506,993	298	5,512x	✓ Covered
NOTCH1	chr9:136,507,285-136,507,457	173	3,059x	✓ Covered
NOTCH1	chr9:136,507,935-136,508,159	225	3,679x	✓ Covered
NOTCH1	chr9:136,508,212-136,508,405	194	5,011x	✓ Covered

FULL PANEL COVERAGE REPORT — Woolf, Virginia A. (NGS-2026-052102)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
NOTCH1	chr9:136,508,850-136,509,091	242	3,796×	✓ Covered
NOTCH1	chr9:136,509,713-136,509,981	269	4,016×	✓ Covered
NOTCH1	chr9:136,510,633-136,510,825	193	3,294×	✓ Covered
NOTCH1	chr9:136,511,132-136,511,291	160	2,851×	✓ Covered
NOTCH1	chr9:136,513,001-136,513,154	154	2,797×	✓ Covered
NOTCH1	chr9:136,513,372-136,513,557	186	3,268×	✓ Covered
NOTCH1	chr9:136,514,490-136,514,722	233	3,746×	✓ Covered
NOTCH1	chr9:136,515,270-136,515,420	151	2,738×	✓ Covered
NOTCH1	chr9:136,515,463-136,515,736	274	5,348×	✓ Covered
NOTCH1	chr9:136,515,961-136,516,114	154	2,820×	✓ Covered
NOTCH1	chr9:136,517,252-136,517,405	154	2,807×	✓ Covered
NOTCH1	chr9:136,517,732-136,517,957	226	3,657×	✓ Covered
NOTCH1	chr9:136,518,117-136,518,312	196	3,606×	✓ Covered
NOTCH1	chr9:136,518,571-136,518,844	274	4,068×	✓ Covered
NOTCH1	chr9:136,519,423-136,519,585	163	2,918×	✓ Covered
NOTCH1	chr9:136,522,830-136,523,208	379	4,590×	✓ Covered
NOTCH1	chr9:136,523,697-136,523,999	303	4,258×	✓ Covered
NOTCH1	chr9:136,544,004-136,544,122	119	2,225×	✓ Covered
NOTCH1	chr9:136,545,706-136,545,806	101	1,941×	✓ Covered

PATIENT & SPECIMEN INFORMATION

Patient name	Dickens, Charles J.	Accession	NGS-2026-052103
Date of birth	1951-09-28 (Age 74)	Specimen	Oral swab, left floor of mouth
MRN	30567843	Collected	2026-05-19
Ordering physician	Dr. Wisdom, DDS MD	Received	2026-05-19
Clinical indication	Oral lesion — left floor of mouth	Reported	2026-05-20

RESULT: VARIANTS DETECTED

DETECTED VARIANTS

Gene	HGVS (coding)	Protein change	VAF	Depth	Tier	Classification
TP53	NM_000546.6:c.524G>A	p.Arg175His	0.60%	5,172x	I	Pathogenic

VARIANT ANNOTATION

TP53 p.Arg175His | Pathogenic (Tier I)

TP53 p.Arg175His is the most frequently observed p53 mutation in human cancer. It disrupts the structural zinc-binding domain of the DNA-binding domain, abolishing wild-type tumour suppressor function, while also conferring dominant-negative and gain-of-function oncogenic properties including promotion of invasion, metastasis, and MDM2 antagonism (COSMIC ID COSM10648). In OSCC, TP53 mutations are present in 50–70% of cases and are associated with aggressive tumour behaviour, locoregional recurrence, and reduced sensitivity to platinum-based chemotherapy. The VAF of 0.60% in a surveillance swab warrants clinical correlation to exclude early local recurrence. Imaging and clinical re-evaluation are recommended.

GENES WITH NO PATHOGENIC OR LIKELY PATHOGENIC VARIANTS DETECTED

AJUBA, CASP8, CDKN2A, CUL3, DPYD, EGFR, EP300, FAT1, FBXW7, HRAS, KEAP1, KMT2D, KRAS, NFE2L2, NOTCH1, NSD1, PIK3CA, PIK3R1, PTEN, TERT
Variants of uncertain significance (Tier III) are not reported.

METHOD SUMMARY

Cell-free and cellular DNA extracted from oral swab is subjected to hybridization capture using the IDT xGen Duplex UMI panel targeting 251 genomic regions (~77 kbp) across 21 cancer-relevant genes. Libraries incorporate 9-bp Unique Molecular Identifiers (UMIs) in a 9M1S+T read structure on both R1 and R2. Libraries are sequenced on Illumina NextSeq (2x150 bp). After alignment to GRCh38 (canonical chromosomes), reads are grouped by UMI using fgbio (adjacency strategy, ≤1 edit distance), and molecular consensus sequences are called requiring ≥2 reads per UMI family. Somatic variants are called on the consensus BAM with VarDict at minimum allele frequency 0.001. Variants ≥0.2% consensus VAF with ≥2 consensus supporting reads are reported. UMI deduplication reduces sequencing noise by >800-fold. Run-level negative controls are included on every sequencing run.

LIMITATIONS

This assay detects somatic single nucleotide variants and small indels within 251 target regions (~77 kbp) at VAF ≥0.2% with validated sensitivity of 98.4% at ≥20,000x raw input depth (~4,000x consensus after UMI deduplication). Structural variants, copy number alterations, gene fusions, large indels, intronic variants outside defined capture regions, and variants in regions with low hybridization capture efficiency are not reliably assessed. A negative result does not exclude malignancy. Tumour heterogeneity, low tumour cell content, and specimen quality may affect variant allele fraction. Variants below the 0.2% VAF reporting threshold may be present but are not reported. At 20,000x raw sequencing depth (~4,000x mean consensus), approximately 0.3% of targets may fall below 1,000x consensus depth; per-target coverage is provided in the laboratory technical report. Results should be interpreted in the context of clinical and pathological findings by a qualified clinician.

Reported by: Dr. Wisdom, MD PhD, FCAP

Report date: 2026-05-20

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

HOTSPOT SCREENING SUMMARY

Patient:

Dickens, Charles J.

Accession:

NGS-2026-052103

Hotspots listed below were explicitly interrogated in addition to all variants above 0.2% VAF with ≥2 consensus reads across all 251 targets. Coverage criteria (≥1,000× consensus depth) were met at all targets.

EGFRCov ✓			HRASCov ✓		
L858R	c.2573T>G	Not detected	G12V	c.35G>T	Not detected
T790M	c.2369C>T	Not detected	G12D	c.35G>A	Not detected
Exon 19 del	c.2235_2249del	Not detected	Q61L	c.182A>T	Not detected
			Q61R	c.182A>G	Not detected
KRASCov ✓			NOTCH1Cov ✓		
G12D	c.35G>T	Not detected	P2415L	c.7244C>T	Not detected
G12V	c.35G>A	Not detected	R1699W	c.5095C>T	Not detected
G12C	c.34G>T	Not detected			
G13D	c.38G>A	Not detected	TP53Cov ✓		
PIK3CACov ✓			R175H	c.524G>A	● DETECTED
E542K	c.1624G>A	Not detected	R248Q	c.743G>A	Not detected
E545K	c.1633G>A	Not detected	R248W	c.742C>T	Not detected
H1047R	c.3140A>G	Not detected	R273H	c.818G>A	Not detected
TERTCov ✓			R282W	c.844C>T	Not detected
C228T	c.-124C>T	Not detected			
C250T	c.-146C>T	Not detected			

REMAINING PANEL GENES — SEQUENCE-LEVEL DETECTION, NO PREDEFINED HOTSPOT LIST

AJUBA · CASP8 · CDKN2A · CUL3 · DPYD · EP300 · FAT1 · FBXW7 · KEAP1 · KMT2D · NFE2L2 · NSD1 · PIK3R1 · PTEN

All variants above 0.2% VAF with ≥2 consensus reads are reported for these genes. No variants meeting reporting criteria were detected. All targets in these genes met minimum coverage criteria (≥1,000× consensus depth).

Reported by: Dr. Wisdom, MD PhD, FCAP

Report date: 2026-05-20

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

FULL PANEL COVERAGE REPORT — Dickens, Charles J. (NGS-2026-052103)

251 targets, 77,417 bp total. Mean consensus depth: 3,663x. Targets below 1,000x consensus: 0 / 251 (0%). Depths are from the UMI-deduplicated consensus BAM (min-reads ≥ 2, 20,000x raw input). Positions reported 1-based.

Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
DPYD				
DPYD	chr1:97,082,291-97,082,491	201	3,441x	✓ Covered
DPYD	chr1:97,449,958-97,450,158	201	3,394x	✓ Covered
DPYD	chr1:97,515,687-97,515,887	201	3,419x	✓ Covered
DPYD	chr1:97,573,763-97,573,963	201	3,375x	✓ Covered
PTEN				
PTEN	chr10:87,864,450-87,864,568	119	2,181x	✓ Covered
PTEN	chr10:87,894,005-87,894,129	125	2,332x	✓ Covered
PTEN	chr10:87,925,493-87,925,577	85	1,632x	✓ Covered
PTEN	chr10:87,931,026-87,931,109	84	1,703x	✓ Covered
PTEN	chr10:87,932,993-87,933,271	279	4,117x	✓ Covered
PTEN	chr10:87,952,098-87,952,279	182	3,193x	✓ Covered
PTEN	chr10:87,957,833-87,958,039	207	3,464x	✓ Covered
PTEN	chr10:87,960,874-87,961,138	265	3,991x	✓ Covered
PTEN	chr10:87,965,267-87,965,489	223	3,485x	✓ Covered
HRAS				
HRAS	chr11:532,619-532,775	157	2,791x	✓ Covered
HRAS	chr11:533,433-533,632	200	3,379x	✓ Covered
HRAS	chr11:533,746-533,964	219	4,213x	✓ Covered
HRAS	chr11:534,192-534,342	151	2,775x	✓ Covered
KRAS				
KRAS	chr12:25,209,778-25,209,931	154	2,776x	✓ Covered
KRAS	chr12:25,225,594-25,225,793	200	3,419x	✓ Covered
KRAS	chr12:25,227,214-25,227,432	219	3,653x	✓ Covered
KRAS	chr12:25,245,254-25,245,404	151	2,756x	✓ Covered
KMT2D				
KMT2D	chr12:49,021,763-49,021,892	130	2,373x	✓ Covered
KMT2D	chr12:49,022,023-49,022,171	149	3,253x	✓ Covered
KMT2D	chr12:49,022,260-49,022,373	114	3,716x	✓ Covered
KMT2D	chr12:49,022,570-49,022,895	326	4,411x	✓ Covered
KMT2D	chr12:49,024,558-49,024,728	171	3,067x	✓ Covered
KMT2D	chr12:49,024,790-49,024,966	177	4,696x	✓ Covered
KMT2D	chr12:49,026,162-49,027,342	1,181	5,772x	✓ Covered
KMT2D	chr12:49,027,783-49,027,950	168	3,044x	✓ Covered
KMT2D	chr12:49,027,989-49,028,161	173	5,141x	✓ Covered
KMT2D	chr12:49,028,808-49,028,978	171	3,032x	✓ Covered
KMT2D	chr12:49,029,041-49,029,256	216	4,841x	✓ Covered
KMT2D	chr12:49,029,381-49,029,496	116	3,096x	✓ Covered
KMT2D	chr12:49,030,260-49,030,459	200	3,421x	✓ Covered
KMT2D	chr12:49,030,581-49,030,788	208	4,076x	✓ Covered
KMT2D	chr12:49,030,873-49,031,053	181	4,348x	✓ Covered
KMT2D	chr12:49,031,155-49,033,984	2,830	6,139x	✓ Covered
KMT2D	chr12:49,034,047-49,034,319	273	5,343x	✓ Covered
KMT2D	chr12:49,034,390-49,034,496	107	4,338x	✓ Covered
KMT2D	chr12:49,034,562-49,034,686	125	4,148x	✓ Covered
KMT2D	chr12:49,034,792-49,034,955	164	3,865x	✓ Covered
KMT2D	chr12:49,037,105-49,039,009	1,905	5,947x	✓ Covered
KMT2D	chr12:49,039,202-49,039,378	177	3,179x	✓ Covered
KMT2D	chr12:49,039,415-49,039,637	223	5,349x	✓ Covered

FULL PANEL COVERAGE REPORT — Dickens, Charles J. (NGS-2026-052103)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
KMT2D	chr12:49,039,704–49,041,555	1,852	6,097×	✓ Covered
KMT2D	chr12:49,041,635–49,041,725	91	4,469×	✓ Covered
KMT2D	chr12:49,041,897–49,042,010	114	2,358×	✓ Covered
KMT2D	chr12:49,042,069–49,042,350	282	5,067×	✓ Covered
KMT2D	chr12:49,042,541–49,042,665	125	2,440×	✓ Covered
KMT2D	chr12:49,042,721–49,042,898	178	4,667×	✓ Covered
KMT2D	chr12:49,043,056–49,043,206	151	3,049×	✓ Covered
KMT2D	chr12:49,043,343–49,043,448	106	2,767×	✓ Covered
KMT2D	chr12:49,043,615–49,043,802	188	3,402×	✓ Covered
KMT2D	chr12:49,043,848–49,044,018	171	5,086×	✓ Covered
KMT2D	chr12:49,044,180–49,044,324	145	2,871×	✓ Covered
KMT2D	chr12:49,044,383–49,044,542	160	4,651×	✓ Covered
KMT2D	chr12:49,044,724–49,044,985	262	4,098×	✓ Covered
KMT2D	chr12:49,045,900–49,045,987	88	1,725×	✓ Covered
KMT2D	chr12:49,046,045–49,046,194	150	4,261×	✓ Covered
KMT2D	chr12:49,046,240–49,046,444	205	5,075×	✓ Covered
KMT2D	chr12:49,046,589–49,046,810	222	3,945×	✓ Covered
KMT2D	chr12:49,047,945–49,048,089	145	2,595×	✓ Covered
KMT2D	chr12:49,048,639–49,048,789	151	2,713×	✓ Covered
KMT2D	chr12:49,049,085–49,049,238	154	2,745×	✓ Covered
KMT2D	chr12:49,049,662–49,050,810	1,149	5,697×	✓ Covered
KMT2D	chr12:49,050,866–49,052,444	1,579	6,142×	✓ Covered
KMT2D	chr12:49,052,544–49,052,729	186	4,347×	✓ Covered
KMT2D	chr12:49,052,895–49,053,092	198	3,609×	✓ Covered
KMT2D	chr12:49,053,187–49,053,341	155	3,843×	✓ Covered
KMT2D	chr12:49,053,456–49,053,661	206	4,113×	✓ Covered
KMT2D	chr12:49,053,958–49,054,160	203	3,424×	✓ Covered
KMT2D	chr12:49,054,287–49,054,436	150	3,346×	✓ Covered
KMT2D	chr12:49,054,508–49,054,771	264	4,873×	✓ Covered
KMT2D	chr12:49,054,880–49,055,046	167	3,936×	✓ Covered
KMT2D	chr12:49,055,256–49,055,344	89	1,842×	✓ Covered
AJUBA				
AJUBA	chr14:22,973,426–22,973,588	163	2,918×	✓ Covered
AJUBA	chr14:22,974,027–22,974,135	109	2,080×	✓ Covered
AJUBA	chr14:22,974,819–22,974,910	92	1,796×	✓ Covered
AJUBA	chr14:22,974,954–22,975,124	171	4,617×	✓ Covered
AJUBA	chr14:22,976,436–22,976,538	103	1,961×	✓ Covered
AJUBA	chr14:22,976,625–22,976,732	108	3,584×	✓ Covered
AJUBA	chr14:22,978,324–22,978,465	142	2,570×	✓ Covered
AJUBA	chr14:22,981,241–22,982,286	1,046	5,679×	✓ Covered
TP53				
TP53	chr17:7,669,592–7,669,710	119	2,163×	✓ Covered
TP53	chr17:7,670,589–7,670,735	147	2,604×	✓ Covered
TP53	chr17:7,673,515–7,673,628	114	2,123×	✓ Covered
TP53	chr17:7,673,681–7,673,857	177	4,707×	✓ Covered
TP53	chr17:7,674,161–7,674,310	150	2,742×	✓ Covered
TP53	chr17:7,674,839–7,674,991	153	2,796×	✓ Covered
TP53	chr17:7,675,033–7,675,256	224	5,172×	● VARIANT DETECTED
TP53	chr17:7,675,974–7,676,292	319	4,346×	✓ Covered
TP53	chr17:7,676,362–7,676,423	62	4,219×	✓ Covered
TP53	chr17:7,676,501–7,676,614	114	3,470×	✓ Covered

FULL PANEL COVERAGE REPORT — Dickens, Charles J. (NGS-2026-052103)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
KEAP1				
KEAP1	chr19:10,486,635–10,486,838	204	3,440×	✓ Covered
KEAP1	chr19:10,489,172–10,489,388	217	3,581×	✓ Covered
KEAP1	chr19:10,489,628–10,489,873	246	3,823×	✓ Covered
KEAP1	chr19:10,491,557–10,492,282	726	5,369×	✓ Covered
KEAP1	chr19:10,499,375–10,500,053	679	5,319×	✓ Covered
NFE2L2				
NFE2L2	chr2:177,233,985–177,234,291	307	4,293×	✓ Covered
CASP8				
CASP8	chr2:201,266,467–201,266,811	345	4,574×	✓ Covered
CASP8	chr2:201,271,496–201,271,641	146	2,674×	✓ Covered
CASP8	chr2:201,272,618–201,272,796	179	3,130×	✓ Covered
CASP8	chr2:201,272,878–201,272,962	85	3,860×	✓ Covered
CASP8	chr2:201,274,869–201,274,973	105	2,000×	✓ Covered
CASP8	chr2:201,276,807–201,276,988	182	3,191×	✓ Covered
CASP8	chr2:201,284,796–201,285,337	542	5,114×	✓ Covered
CASP8	chr2:201,286,439–201,286,611	173	3,039×	✓ Covered
CUL3				
CUL3	chr2:224,474,228–224,474,396	169	2,981×	✓ Covered
CUL3	chr2:224,478,180–224,478,365	186	3,208×	✓ Covered
CUL3	chr2:224,481,872–224,482,098	227	3,699×	✓ Covered
CUL3	chr2:224,495,812–224,495,986	175	3,032×	✓ Covered
CUL3	chr2:224,497,733–224,497,869	137	2,522×	✓ Covered
CUL3	chr2:224,500,343–224,500,507	165	2,929×	✓ Covered
CUL3	chr2:224,502,945–224,503,092	148	2,675×	✓ Covered
CUL3	chr2:224,503,632–224,503,842	211	3,468×	✓ Covered
CUL3	chr2:224,505,936–224,506,152	217	3,641×	✓ Covered
CUL3	chr2:224,506,838–224,507,023	186	3,226×	✓ Covered
CUL3	chr2:224,511,334–224,511,602	269	4,036×	✓ Covered
CUL3	chr2:224,513,504–224,513,658	155	2,773×	✓ Covered
CUL3	chr2:224,514,592–224,514,792	201	3,396×	✓ Covered
CUL3	chr2:224,535,508–224,535,661	154	2,726×	✓ Covered
CUL3	chr2:224,557,639–224,557,876	238	3,775×	✓ Covered
CUL3	chr2:224,584,924–224,585,029	106	2,054×	✓ Covered
EP300				
EP300	chr22:41,092,985–41,093,118	134	2,472×	✓ Covered
EP300	chr22:41,117,167–41,117,841	675	5,300×	✓ Covered
EP300	chr22:41,125,844–41,126,060	217	3,588×	✓ Covered
EP300	chr22:41,127,467–41,127,768	302	4,289×	✓ Covered
EP300	chr22:41,129,870–41,130,023	154	2,809×	✓ Covered
EP300	chr22:41,131,368–41,131,653	286	4,195×	✓ Covered
EP300	chr22:41,135,793–41,135,926	134	2,474×	✓ Covered
EP300	chr22:41,137,633–41,137,810	178	3,102×	✓ Covered
EP300	chr22:41,140,120–41,140,277	158	2,823×	✓ Covered
EP300	chr22:41,141,028–41,141,242	215	3,591×	✓ Covered
EP300	chr22:41,146,719–41,146,836	118	2,198×	✓ Covered
EP300	chr22:41,147,817–41,147,966	150	2,784×	✓ Covered
EP300	chr22:41,149,018–41,149,195	178	3,178×	✓ Covered
EP300	chr22:41,149,741–41,150,218	478	4,937×	✓ Covered
EP300	chr22:41,151,813–41,152,032	220	3,626×	✓ Covered
EP300	chr22:41,152,186–41,152,370	185	3,516×	✓ Covered

FULL PANEL COVERAGE REPORT — Dickens, Charles J. (NGS-2026-052103)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
EP300	chr22:41,154,975-41,155,133	159	2,847×	✓ Covered
EP300	chr22:41,157,149-41,157,428	280	4,114×	✓ Covered
EP300	chr22:41,158,392-41,158,520	129	2,377×	✓ Covered
EP300	chr22:41,160,622-41,160,742	121	2,250×	✓ Covered
EP300	chr22:41,162,703-41,162,799	97	1,842×	✓ Covered
EP300	chr22:41,164,033-41,164,150	118	2,181×	✓ Covered
EP300	chr22:41,166,579-41,166,686	108	2,056×	✓ Covered
EP300	chr22:41,168,429-41,168,619	191	3,290×	✓ Covered
EP300	chr22:41,168,701-41,168,887	187	4,447×	✓ Covered
EP300	chr22:41,169,483-41,169,636	154	2,701×	✓ Covered
EP300	chr22:41,170,386-41,170,591	206	3,435×	✓ Covered
EP300	chr22:41,172,479-41,172,683	205	3,458×	✓ Covered
EP300	chr22:41,173,603-41,173,804	202	3,382×	✓ Covered
EP300	chr22:41,176,227-41,176,548	322	4,334×	✓ Covered
EP300	chr22:41,176,753-41,178,973	2,221	6,007×	✓ Covered
PIK3CA				
PIK3CA	chr3:179,218,190-179,218,354	165	2,906×	✓ Covered
PIK3CA	chr3:179,234,074-179,234,381	308	4,231×	✓ Covered
FBXW7				
FBXW7	chr4:152,325,986-152,326,251	266	4,075×	✓ Covered
FAT1				
FAT1	chr4:186,588,575-186,589,240	666	5,274×	✓ Covered
FAT1	chr4:186,595,669-186,595,846	178	3,144×	✓ Covered
FAT1	chr4:186,596,520-186,597,191	672	5,324×	✓ Covered
FAT1	chr4:186,597,662-186,597,812	151	2,746×	✓ Covered
FAT1	chr4:186,597,952-186,598,145	194	3,700×	✓ Covered
FAT1	chr4:186,599,878-186,600,380	503	5,011×	✓ Covered
FAT1	chr4:186,601,249-186,601,446	198	3,351×	✓ Covered
FAT1	chr4:186,602,883-186,603,054	172	3,033×	✓ Covered
FAT1	chr4:186,603,156-186,603,997	842	5,724×	✓ Covered
FAT1	chr4:186,604,357-186,604,594	238	3,808×	✓ Covered
FAT1	chr4:186,606,050-186,606,233	184	3,248×	✓ Covered
FAT1	chr4:186,609,163-186,609,340	178	3,179×	✓ Covered
FAT1	chr4:186,609,781-186,610,035	255	3,859×	✓ Covered
FAT1	chr4:186,611,366-186,611,795	430	4,789×	✓ Covered
FAT1	chr4:186,613,089-186,613,362	274	4,091×	✓ Covered
FAT1	chr4:186,614,171-186,614,364	194	3,310×	✓ Covered
FAT1	chr4:186,616,985-186,617,221	237	3,759×	✓ Covered
FAT1	chr4:186,617,688-186,621,795	4,108	6,152×	✓ Covered
FAT1	chr4:186,628,134-186,628,384	251	3,863×	✓ Covered
FAT1	chr4:186,628,468-186,628,783	316	5,042×	✓ Covered
FAT1	chr4:186,633,664-186,633,843	180	3,106×	✓ Covered
FAT1	chr4:186,636,005-186,636,255	251	3,915×	✓ Covered
FAT1	chr4:186,636,565-186,636,934	370	4,616×	✓ Covered
FAT1	chr4:186,639,702-186,639,803	102	1,982×	✓ Covered
FAT1	chr4:186,663,279-186,663,633	355	4,544×	✓ Covered
FAT1	chr4:186,706,543-186,709,847	3,305	6,105×	✓ Covered
TERT				
TERT	chr5:1,295,050-1,295,250	201	3,413×	✓ Covered
PIK3R1				
PIK3R1	chr5:68,294,516-68,294,698	183	3,114×	✓ Covered

FULL PANEL COVERAGE REPORT — Dickens, Charles J. (NGS-2026-052103)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
PIK3R1	chr5:68,295,128-68,295,344	217	3,576x	✓ Covered
PIK3R1	chr5:68,295,400-68,295,508	109	4,634x	✓ Covered
NSD1				
NSD1	chr5:177,135,084-177,136,050	967	5,592x	✓ Covered
NSD1	chr5:177,191,864-177,192,039	176	3,083x	✓ Covered
NSD1	chr5:177,204,100-177,204,312	213	3,560x	✓ Covered
NSD1	chr5:177,209,616-177,212,215	2,600	6,056x	✓ Covered
NSD1	chr5:177,235,801-177,235,965	165	3,008x	✓ Covered
NSD1	chr5:177,238,217-177,238,527	311	4,342x	✓ Covered
NSD1	chr5:177,239,736-177,239,885	150	2,702x	✓ Covered
NSD1	chr5:177,244,175-177,244,290	116	2,161x	✓ Covered
NSD1	chr5:177,246,658-177,246,816	159	2,894x	✓ Covered
NSD1	chr5:177,248,161-177,248,344	184	3,201x	✓ Covered
NSD1	chr5:177,251,710-177,251,873	164	2,901x	✓ Covered
NSD1	chr5:177,256,931-177,257,171	241	3,820x	✓ Covered
NSD1	chr5:177,259,969-177,260,188	220	3,595x	✓ Covered
NSD1	chr5:177,267,542-177,267,738	197	3,375x	✓ Covered
NSD1	chr5:177,269,582-177,269,827	246	3,893x	✓ Covered
NSD1	chr5:177,273,652-177,273,804	153	2,755x	✓ Covered
NSD1	chr5:177,280,545-177,280,854	310	4,326x	✓ Covered
NSD1	chr5:177,282,445-177,282,601	157	2,840x	✓ Covered
NSD1	chr5:177,283,767-177,283,948	182	3,142x	✓ Covered
NSD1	chr5:177,288,799-177,288,945	147	2,700x	✓ Covered
NSD1	chr5:177,291,934-177,292,178	245	3,838x	✓ Covered
NSD1	chr5:177,293,812-177,295,476	1,665	5,916x	✓ Covered
EGFR				
EGFR	chr7:55,173,901-55,174,063	163	2,862x	✓ Covered
EGFR	chr7:55,174,702-55,174,840	139	2,522x	✓ Covered
EGFR	chr7:55,181,273-55,181,498	226	3,680x	✓ Covered
EGFR	chr7:55,191,699-55,191,894	196	3,285x	✓ Covered
CDKN2A				
CDKN2A	chr9:21,968,212-21,968,262	51	1,071x	✓ Covered
CDKN2A	chr9:21,970,882-21,971,228	347	4,497x	✓ Covered
CDKN2A	chr9:21,974,658-21,974,846	189	6,567x	✓ Covered
CDKN2A	chr9:21,974,658-21,974,847	190	6,585x	✓ Covered
NOTCH1				
NOTCH1	chr9:136,496,054-136,497,578	1,525	5,905x	✓ Covered
NOTCH1	chr9:136,498,879-136,499,016	138	2,558x	✓ Covered
NOTCH1	chr9:136,499,092-136,499,279	188	4,499x	✓ Covered
NOTCH1	chr9:136,500,532-136,500,867	336	4,442x	✓ Covered
NOTCH1	chr9:136,501,728-136,501,933	206	3,512x	✓ Covered
NOTCH1	chr9:136,501,981-136,502,108	128	4,834x	✓ Covered
NOTCH1	chr9:136,502,252-136,502,508	257	4,169x	✓ Covered
NOTCH1	chr9:136,503,162-136,503,350	189	3,223x	✓ Covered
NOTCH1	chr9:136,504,653-136,505,124	472	4,922x	✓ Covered
NOTCH1	chr9:136,505,290-136,505,901	612	5,328x	✓ Covered
NOTCH1	chr9:136,506,507-136,506,659	153	2,798x	✓ Covered
NOTCH1	chr9:136,506,696-136,506,993	298	5,441x	✓ Covered
NOTCH1	chr9:136,507,285-136,507,457	173	3,014x	✓ Covered
NOTCH1	chr9:136,507,935-136,508,159	225	3,693x	✓ Covered
NOTCH1	chr9:136,508,212-136,508,405	194	5,012x	✓ Covered

FULL PANEL COVERAGE REPORT — Dickens, Charles J. (NGS-2026-052103)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
NOTCH1	chr9:136,508,850-136,509,091	242	3,810×	✓ Covered
NOTCH1	chr9:136,509,713-136,509,981	269	4,098×	✓ Covered
NOTCH1	chr9:136,510,633-136,510,825	193	3,344×	✓ Covered
NOTCH1	chr9:136,511,132-136,511,291	160	2,888×	✓ Covered
NOTCH1	chr9:136,513,001-136,513,154	154	2,750×	✓ Covered
NOTCH1	chr9:136,513,372-136,513,557	186	3,211×	✓ Covered
NOTCH1	chr9:136,514,490-136,514,722	233	3,773×	✓ Covered
NOTCH1	chr9:136,515,270-136,515,420	151	2,680×	✓ Covered
NOTCH1	chr9:136,515,463-136,515,736	274	5,344×	✓ Covered
NOTCH1	chr9:136,515,961-136,516,114	154	2,799×	✓ Covered
NOTCH1	chr9:136,517,252-136,517,405	154	2,762×	✓ Covered
NOTCH1	chr9:136,517,732-136,517,957	226	3,636×	✓ Covered
NOTCH1	chr9:136,518,117-136,518,312	196	3,595×	✓ Covered
NOTCH1	chr9:136,518,571-136,518,844	274	4,115×	✓ Covered
NOTCH1	chr9:136,519,423-136,519,585	163	2,925×	✓ Covered
NOTCH1	chr9:136,522,830-136,523,208	379	4,553×	✓ Covered
NOTCH1	chr9:136,523,697-136,523,999	303	4,203×	✓ Covered
NOTCH1	chr9:136,544,004-136,544,122	119	2,275×	✓ Covered
NOTCH1	chr9:136,545,706-136,545,806	101	1,935×	✓ Covered

PATIENT & SPECIMEN INFORMATION			
Patient name	Kafka, Franz J.	Accession	NGS-2026-052105
Date of birth	1980-02-17 (Age 46)	Specimen	Oral swab, anterior tongue
MRN	50789065	Collected	2026-05-19
Ordering physician	Dr. Wisdom, MD	Received	2026-05-19
Clinical indication	Oral lesion — anterior tongue (surveillance post-resection)		2026-05-20

RESULT: NO REPORTABLE VARIANTS DETECTED

INTERPRETATION

No pathogenic or likely pathogenic somatic variants were detected in the 21 genes covered by this assay at $\geq 0.2\%$ VAF with ≥ 2 consensus supporting reads. A negative result does not exclude malignancy. Variants below the reporting threshold, variants outside the 251 assay target regions, structural variants, and copy number alterations are not assessed by this assay. Clinical correlation is required; repeat testing or biopsy should be considered if clinical suspicion persists.

GENES WITH NO PATHOGENIC OR LIKELY PATHOGENIC VARIANTS DETECTED

AJUBA, CASP8, CDKN2A, CUL3, DPYD, EGFR, EP300, FAT1, FBXW7, HRAS, KEAP1, KMT2D, KRAS, NFE2L2, NOTCH1, NSD1, PIK3CA, PIK3R1, PTEN, TERT, TP53

Variants of uncertain significance (Tier III) are not reported.

METHOD SUMMARY

Cell-free and cellular DNA extracted from oral swab is subjected to hybridization capture using the IDT xGen Duplex UMI panel targeting 251 genomic regions (~77 kbp) across 21 cancer-relevant genes. Libraries incorporate 9-bp Unique Molecular Identifiers (UMIs) in a 9M1S+T read structure on both R1 and R2. Libraries are sequenced on Illumina NextSeq (2x150 bp). After alignment to GRCh38 (canonical chromosomes), reads are grouped by UMI using fgbio (adjacency strategy, ≤ 1 edit distance), and molecular consensus sequences are called requiring ≥ 2 reads per UMI family. Somatic variants are called on the consensus BAM with VarDict at minimum allele frequency 0.001. Variants $\geq 0.2\%$ consensus VAF with ≥ 2 consensus supporting reads are reported. UMI deduplication reduces sequencing noise by >800 -fold. Run-level negative controls are included on every sequencing run.

LIMITATIONS

This assay detects somatic single nucleotide variants and small indels within 251 target regions (~77 kbp) at VAF $\geq 0.2\%$ with validated sensitivity of 98.4% at $\geq 20,000\times$ raw input depth (~4,000 $\times$ consensus after UMI deduplication). Structural variants, copy number alterations, gene fusions, large indels, intronic variants outside defined capture regions, and variants in regions with low hybridization capture efficiency are not reliably assessed. A negative result does not exclude malignancy. Tumour heterogeneity, low tumour cell content, and specimen quality may affect variant allele fraction. Variants below the 0.2% VAF reporting threshold may be present but are not reported. At 20,000 $\times$ raw sequencing depth (~4,000 $\times$ mean consensus), approximately 0.3% of targets may fall below 1,000 $\times$ consensus depth; per-target coverage is provided in the laboratory technical report. Results should be interpreted in the context of clinical and pathological findings by a qualified clinician.

Reported by: Dr. Wisdom, MD PhD, FCAP

Report date: 2026-05-20

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

HOTSPOT SCREENING SUMMARY

Patient: Kafka, Franz J.

Accession: NGS-2026-052105

Hotspots listed below were explicitly interrogated in addition to all variants above 0.2% VAF with ≥2 consensus reads across all 251 targets. Coverage criteria (≥1,000× consensus depth) were met at all targets.

EGFRCov ✓			HRASCov ✓		
L858R	c.2573T>G	Not detected	G12V	c.35G>T	Not detected
T790M	c.2369C>T	Not detected	G12D	c.35G>A	Not detected
Exon 19 del	c.2235_2249del	Not detected	Q61L	c.182A>T	Not detected
			Q61R	c.182A>G	Not detected
KRASCov ✓			NOTCH1Cov ✓		
G12D	c.35G>T	Not detected	P2415L	c.7244C>T	Not detected
G12V	c.35G>A	Not detected	R1699W	c.5095C>T	Not detected
G12C	c.34G>T	Not detected			
G13D	c.38G>A	Not detected	TP53Cov ✓		
PIK3CACov ✓			R175H	c.524G>A	Not detected
E542K	c.1624G>A	Not detected	R248Q	c.743G>A	Not detected
E545K	c.1633G>A	Not detected	R248W	c.742C>T	Not detected
H1047R	c.3140A>G	Not detected	R273H	c.818G>A	Not detected
TERTCov ✓			R282W	c.844C>T	Not detected
C228T	c.-124C>T	Not detected			
C250T	c.-146C>T	Not detected			

REMAINING PANEL GENES — SEQUENCE-LEVEL DETECTION, NO PREDEFINED HOTSPOT LIST

AJUBA · CASP8 · CDKN2A · CUL3 · DPYD · EP300 · FAT1 · FBXW7 · KEAP1 · KMT2D · NFE2L2 · NSD1 · PIK3R1 · PTEN

All variants above 0.2% VAF with ≥2 consensus reads are reported for these genes. No variants meeting reporting criteria were detected. All targets in these genes met minimum coverage criteria (≥1,000× consensus depth).

Reported by: Dr. Wisdom, MD PhD, FCAP

Report date: 2026-05-20

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

FULL PANEL COVERAGE REPORT — Kafka, Franz J. (NGS-2026-052105)

251 targets, 77,417 bp total. Mean consensus depth: 3,667x. Targets below 1,000x consensus: 0 / 251 (0%). Depths are from the UMI-deduplicated consensus BAM (min-reads ≥ 2, 20,000x raw input). Positions reported 1-based.

Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
DPYD				
DPYD	chr1:97,082,291-97,082,491	201	3,381x	✓ Covered
DPYD	chr1:97,449,958-97,450,158	201	3,440x	✓ Covered
DPYD	chr1:97,515,687-97,515,887	201	3,406x	✓ Covered
DPYD	chr1:97,573,763-97,573,963	201	3,387x	✓ Covered
PTEN				
PTEN	chr10:87,864,450-87,864,568	119	2,281x	✓ Covered
PTEN	chr10:87,894,005-87,894,129	125	2,316x	✓ Covered
PTEN	chr10:87,925,493-87,925,577	85	1,708x	✓ Covered
PTEN	chr10:87,931,026-87,931,109	84	1,688x	✓ Covered
PTEN	chr10:87,932,993-87,933,271	279	4,096x	✓ Covered
PTEN	chr10:87,952,098-87,952,279	182	3,198x	✓ Covered
PTEN	chr10:87,957,833-87,958,039	207	3,479x	✓ Covered
PTEN	chr10:87,960,874-87,961,138	265	4,054x	✓ Covered
PTEN	chr10:87,965,267-87,965,489	223	3,473x	✓ Covered
HRAS				
HRAS	chr11:532,619-532,775	157	2,847x	✓ Covered
HRAS	chr11:533,433-533,632	200	3,381x	✓ Covered
HRAS	chr11:533,746-533,964	219	4,273x	✓ Covered
HRAS	chr11:534,192-534,342	151	2,775x	✓ Covered
KRAS				
KRAS	chr12:25,209,778-25,209,931	154	2,803x	✓ Covered
KRAS	chr12:25,225,594-25,225,793	200	3,376x	✓ Covered
KRAS	chr12:25,227,214-25,227,432	219	3,621x	✓ Covered
KRAS	chr12:25,245,254-25,245,404	151	2,778x	✓ Covered
KMT2D				
KMT2D	chr12:49,021,763-49,021,892	130	2,406x	✓ Covered
KMT2D	chr12:49,022,023-49,022,171	149	3,294x	✓ Covered
KMT2D	chr12:49,022,260-49,022,373	114	3,752x	✓ Covered
KMT2D	chr12:49,022,570-49,022,895	326	4,376x	✓ Covered
KMT2D	chr12:49,024,558-49,024,728	171	3,054x	✓ Covered
KMT2D	chr12:49,024,790-49,024,966	177	4,660x	✓ Covered
KMT2D	chr12:49,026,162-49,027,342	1,181	5,754x	✓ Covered
KMT2D	chr12:49,027,783-49,027,950	168	2,973x	✓ Covered
KMT2D	chr12:49,027,989-49,028,161	173	5,123x	✓ Covered
KMT2D	chr12:49,028,808-49,028,978	171	3,050x	✓ Covered
KMT2D	chr12:49,029,041-49,029,256	216	4,842x	✓ Covered
KMT2D	chr12:49,029,381-49,029,496	116	3,088x	✓ Covered
KMT2D	chr12:49,030,260-49,030,459	200	3,443x	✓ Covered
KMT2D	chr12:49,030,581-49,030,788	208	4,055x	✓ Covered
KMT2D	chr12:49,030,873-49,031,053	181	4,336x	✓ Covered
KMT2D	chr12:49,031,155-49,033,984	2,830	6,125x	✓ Covered
KMT2D	chr12:49,034,047-49,034,319	273	5,327x	✓ Covered
KMT2D	chr12:49,034,390-49,034,496	107	4,350x	✓ Covered
KMT2D	chr12:49,034,562-49,034,686	125	4,110x	✓ Covered
KMT2D	chr12:49,034,792-49,034,955	164	3,766x	✓ Covered
KMT2D	chr12:49,037,105-49,039,009	1,905	5,939x	✓ Covered
KMT2D	chr12:49,039,202-49,039,378	177	3,211x	✓ Covered
KMT2D	chr12:49,039,415-49,039,637	223	5,323x	✓ Covered

FULL PANEL COVERAGE REPORT — Kafka, Franz J. (NGS-2026-052105)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
KMT2D	chr12:49,039,704–49,041,555	1,852	6,094×	✓ Covered
KMT2D	chr12:49,041,635–49,041,725	91	4,502×	✓ Covered
KMT2D	chr12:49,041,897–49,042,010	114	2,367×	✓ Covered
KMT2D	chr12:49,042,069–49,042,350	282	5,059×	✓ Covered
KMT2D	chr12:49,042,541–49,042,665	125	2,457×	✓ Covered
KMT2D	chr12:49,042,721–49,042,898	178	4,739×	✓ Covered
KMT2D	chr12:49,043,056–49,043,206	151	3,045×	✓ Covered
KMT2D	chr12:49,043,343–49,043,448	106	2,786×	✓ Covered
KMT2D	chr12:49,043,615–49,043,802	188	3,413×	✓ Covered
KMT2D	chr12:49,043,848–49,044,018	171	5,012×	✓ Covered
KMT2D	chr12:49,044,180–49,044,324	145	2,863×	✓ Covered
KMT2D	chr12:49,044,383–49,044,542	160	4,617×	✓ Covered
KMT2D	chr12:49,044,724–49,044,985	262	4,062×	✓ Covered
KMT2D	chr12:49,045,900–49,045,987	88	1,704×	✓ Covered
KMT2D	chr12:49,046,045–49,046,194	150	4,280×	✓ Covered
KMT2D	chr12:49,046,240–49,046,444	205	5,105×	✓ Covered
KMT2D	chr12:49,046,589–49,046,810	222	3,937×	✓ Covered
KMT2D	chr12:49,047,945–49,048,089	145	2,646×	✓ Covered
KMT2D	chr12:49,048,639–49,048,789	151	2,707×	✓ Covered
KMT2D	chr12:49,049,085–49,049,238	154	2,742×	✓ Covered
KMT2D	chr12:49,049,662–49,050,810	1,149	5,752×	✓ Covered
KMT2D	chr12:49,050,866–49,052,444	1,579	6,143×	✓ Covered
KMT2D	chr12:49,052,544–49,052,729	186	4,347×	✓ Covered
KMT2D	chr12:49,052,895–49,053,092	198	3,569×	✓ Covered
KMT2D	chr12:49,053,187–49,053,341	155	3,943×	✓ Covered
KMT2D	chr12:49,053,456–49,053,661	206	4,053×	✓ Covered
KMT2D	chr12:49,053,958–49,054,160	203	3,430×	✓ Covered
KMT2D	chr12:49,054,287–49,054,436	150	3,409×	✓ Covered
KMT2D	chr12:49,054,508–49,054,771	264	4,923×	✓ Covered
KMT2D	chr12:49,054,880–49,055,046	167	3,851×	✓ Covered
KMT2D	chr12:49,055,256–49,055,344	89	1,789×	✓ Covered
AJUBA				
AJUBA	chr14:22,973,426–22,973,588	163	2,914×	✓ Covered
AJUBA	chr14:22,974,027–22,974,135	109	2,083×	✓ Covered
AJUBA	chr14:22,974,819–22,974,910	92	1,794×	✓ Covered
AJUBA	chr14:22,974,954–22,975,124	171	4,638×	✓ Covered
AJUBA	chr14:22,976,436–22,976,538	103	1,970×	✓ Covered
AJUBA	chr14:22,976,625–22,976,732	108	3,547×	✓ Covered
AJUBA	chr14:22,978,324–22,978,465	142	2,578×	✓ Covered
AJUBA	chr14:22,981,241–22,982,286	1,046	5,639×	✓ Covered
TP53				
TP53	chr17:7,669,592–7,669,710	119	2,190×	✓ Covered
TP53	chr17:7,670,589–7,670,735	147	2,707×	✓ Covered
TP53	chr17:7,673,515–7,673,628	114	2,145×	✓ Covered
TP53	chr17:7,673,681–7,673,857	177	4,767×	✓ Covered
TP53	chr17:7,674,161–7,674,310	150	2,716×	✓ Covered
TP53	chr17:7,674,839–7,674,991	153	2,742×	✓ Covered
TP53	chr17:7,675,033–7,675,256	224	5,251×	✓ Covered
TP53	chr17:7,675,974–7,676,292	319	4,359×	✓ Covered
TP53	chr17:7,676,362–7,676,423	62	4,280×	✓ Covered
TP53	chr17:7,676,501–7,676,614	114	3,510×	✓ Covered

FULL PANEL COVERAGE REPORT — Kafka, Franz J. (NGS-2026-052105)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
KEAP1				
KEAP1	chr19:10,486,635–10,486,838	204	3,429×	✓ Covered
KEAP1	chr19:10,489,172–10,489,388	217	3,629×	✓ Covered
KEAP1	chr19:10,489,628–10,489,873	246	3,809×	✓ Covered
KEAP1	chr19:10,491,557–10,492,282	726	5,387×	✓ Covered
KEAP1	chr19:10,499,375–10,500,053	679	5,320×	✓ Covered
NFE2L2				
NFE2L2	chr2:177,233,985–177,234,291	307	4,315×	✓ Covered
CASP8				
CASP8	chr2:201,266,467–201,266,811	345	4,467×	✓ Covered
CASP8	chr2:201,271,496–201,271,641	146	2,691×	✓ Covered
CASP8	chr2:201,272,618–201,272,796	179	3,133×	✓ Covered
CASP8	chr2:201,272,878–201,272,962	85	3,805×	✓ Covered
CASP8	chr2:201,274,869–201,274,973	105	2,020×	✓ Covered
CASP8	chr2:201,276,807–201,276,988	182	3,164×	✓ Covered
CASP8	chr2:201,284,796–201,285,337	542	5,134×	✓ Covered
CASP8	chr2:201,286,439–201,286,611	173	3,048×	✓ Covered
CUL3				
CUL3	chr2:224,474,228–224,474,396	169	2,999×	✓ Covered
CUL3	chr2:224,478,180–224,478,365	186	3,215×	✓ Covered
CUL3	chr2:224,481,872–224,482,098	227	3,736×	✓ Covered
CUL3	chr2:224,495,812–224,495,986	175	3,105×	✓ Covered
CUL3	chr2:224,497,733–224,497,869	137	2,600×	✓ Covered
CUL3	chr2:224,500,343–224,500,507	165	2,955×	✓ Covered
CUL3	chr2:224,502,945–224,503,092	148	2,653×	✓ Covered
CUL3	chr2:224,503,632–224,503,842	211	3,490×	✓ Covered
CUL3	chr2:224,505,936–224,506,152	217	3,612×	✓ Covered
CUL3	chr2:224,506,838–224,507,023	186	3,282×	✓ Covered
CUL3	chr2:224,511,334–224,511,602	269	4,062×	✓ Covered
CUL3	chr2:224,513,504–224,513,658	155	2,799×	✓ Covered
CUL3	chr2:224,514,592–224,514,792	201	3,457×	✓ Covered
CUL3	chr2:224,535,508–224,535,661	154	2,806×	✓ Covered
CUL3	chr2:224,557,639–224,557,876	238	3,767×	✓ Covered
CUL3	chr2:224,584,924–224,585,029	106	2,026×	✓ Covered
EP300				
EP300	chr22:41,092,985–41,093,118	134	2,456×	✓ Covered
EP300	chr22:41,117,167–41,117,841	675	5,355×	✓ Covered
EP300	chr22:41,125,844–41,126,060	217	3,609×	✓ Covered
EP300	chr22:41,127,467–41,127,768	302	4,248×	✓ Covered
EP300	chr22:41,129,870–41,130,023	154	2,698×	✓ Covered
EP300	chr22:41,131,368–41,131,653	286	4,174×	✓ Covered
EP300	chr22:41,135,793–41,135,926	134	2,462×	✓ Covered
EP300	chr22:41,137,633–41,137,810	178	3,057×	✓ Covered
EP300	chr22:41,140,120–41,140,277	158	2,870×	✓ Covered
EP300	chr22:41,141,028–41,141,242	215	3,576×	✓ Covered
EP300	chr22:41,146,719–41,146,836	118	2,269×	✓ Covered
EP300	chr22:41,147,817–41,147,966	150	2,697×	✓ Covered
EP300	chr22:41,149,018–41,149,195	178	3,092×	✓ Covered
EP300	chr22:41,149,741–41,150,218	478	4,994×	✓ Covered
EP300	chr22:41,151,813–41,152,032	220	3,671×	✓ Covered
EP300	chr22:41,152,186–41,152,370	185	3,449×	✓ Covered

FULL PANEL COVERAGE REPORT — Kafka, Franz J. (NGS-2026-052105)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
EP300	chr22:41,154,975-41,155,133	159	2,866×	✓ Covered
EP300	chr22:41,157,149-41,157,428	280	4,100×	✓ Covered
EP300	chr22:41,158,392-41,158,520	129	2,426×	✓ Covered
EP300	chr22:41,160,622-41,160,742	121	2,226×	✓ Covered
EP300	chr22:41,162,703-41,162,799	97	1,848×	✓ Covered
EP300	chr22:41,164,033-41,164,150	118	2,178×	✓ Covered
EP300	chr22:41,166,579-41,166,686	108	2,036×	✓ Covered
EP300	chr22:41,168,429-41,168,619	191	3,303×	✓ Covered
EP300	chr22:41,168,701-41,168,887	187	4,329×	✓ Covered
EP300	chr22:41,169,483-41,169,636	154	2,788×	✓ Covered
EP300	chr22:41,170,386-41,170,591	206	3,462×	✓ Covered
EP300	chr22:41,172,479-41,172,683	205	3,519×	✓ Covered
EP300	chr22:41,173,603-41,173,804	202	3,393×	✓ Covered
EP300	chr22:41,176,227-41,176,548	322	4,381×	✓ Covered
EP300	chr22:41,176,753-41,178,973	2,221	6,007×	✓ Covered
PIK3CA				
PIK3CA	chr3:179,218,190-179,218,354	165	2,923×	✓ Covered
PIK3CA	chr3:179,234,074-179,234,381	308	4,293×	✓ Covered
FBXW7				
FBXW7	chr4:152,325,986-152,326,251	266	3,982×	✓ Covered
FAT1				
FAT1	chr4:186,588,575-186,589,240	666	5,312×	✓ Covered
FAT1	chr4:186,595,669-186,595,846	178	3,160×	✓ Covered
FAT1	chr4:186,596,520-186,597,191	672	5,308×	✓ Covered
FAT1	chr4:186,597,662-186,597,812	151	2,692×	✓ Covered
FAT1	chr4:186,597,952-186,598,145	194	3,716×	✓ Covered
FAT1	chr4:186,599,878-186,600,380	503	5,062×	✓ Covered
FAT1	chr4:186,601,249-186,601,446	198	3,329×	✓ Covered
FAT1	chr4:186,602,883-186,603,054	172	3,012×	✓ Covered
FAT1	chr4:186,603,156-186,603,997	842	5,716×	✓ Covered
FAT1	chr4:186,604,357-186,604,594	238	3,816×	✓ Covered
FAT1	chr4:186,606,050-186,606,233	184	3,265×	✓ Covered
FAT1	chr4:186,609,163-186,609,340	178	3,130×	✓ Covered
FAT1	chr4:186,609,781-186,610,035	255	3,973×	✓ Covered
FAT1	chr4:186,611,366-186,611,795	430	4,815×	✓ Covered
FAT1	chr4:186,613,089-186,613,362	274	4,070×	✓ Covered
FAT1	chr4:186,614,171-186,614,364	194	3,336×	✓ Covered
FAT1	chr4:186,616,985-186,617,221	237	3,784×	✓ Covered
FAT1	chr4:186,617,688-186,621,795	4,108	6,164×	✓ Covered
FAT1	chr4:186,628,134-186,628,384	251	3,898×	✓ Covered
FAT1	chr4:186,628,468-186,628,783	316	5,081×	✓ Covered
FAT1	chr4:186,633,664-186,633,843	180	3,154×	✓ Covered
FAT1	chr4:186,636,005-186,636,255	251	3,906×	✓ Covered
FAT1	chr4:186,636,565-186,636,934	370	4,569×	✓ Covered
FAT1	chr4:186,639,702-186,639,803	102	1,967×	✓ Covered
FAT1	chr4:186,663,279-186,663,633	355	4,506×	✓ Covered
FAT1	chr4:186,706,543-186,709,847	3,305	6,128×	✓ Covered
TERT				
TERT	chr5:1,295,050-1,295,250	201	3,422×	✓ Covered
PIK3R1				
PIK3R1	chr5:68,294,516-68,294,698	183	3,230×	✓ Covered

FULL PANEL COVERAGE REPORT — Kafka, Franz J. (NGS-2026-052105)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
PIK3R1	chr5:68,295,128-68,295,344	217	3,604x	✓ Covered
PIK3R1	chr5:68,295,400-68,295,508	109	4,680x	✓ Covered
NSD1				
NSD1	chr5:177,135,084-177,136,050	967	5,606x	✓ Covered
NSD1	chr5:177,191,864-177,192,039	176	3,054x	✓ Covered
NSD1	chr5:177,204,100-177,204,312	213	3,526x	✓ Covered
NSD1	chr5:177,209,616-177,212,215	2,600	6,068x	✓ Covered
NSD1	chr5:177,235,801-177,235,965	165	2,957x	✓ Covered
NSD1	chr5:177,238,217-177,238,527	311	4,359x	✓ Covered
NSD1	chr5:177,239,736-177,239,885	150	2,704x	✓ Covered
NSD1	chr5:177,244,175-177,244,290	116	2,204x	✓ Covered
NSD1	chr5:177,246,658-177,246,816	159	2,858x	✓ Covered
NSD1	chr5:177,248,161-177,248,344	184	3,176x	✓ Covered
NSD1	chr5:177,251,710-177,251,873	164	2,904x	✓ Covered
NSD1	chr5:177,256,931-177,257,171	241	3,832x	✓ Covered
NSD1	chr5:177,259,969-177,260,188	220	3,613x	✓ Covered
NSD1	chr5:177,267,542-177,267,738	197	3,354x	✓ Covered
NSD1	chr5:177,269,582-177,269,827	246	3,868x	✓ Covered
NSD1	chr5:177,273,652-177,273,804	153	2,775x	✓ Covered
NSD1	chr5:177,280,545-177,280,854	310	4,324x	✓ Covered
NSD1	chr5:177,282,445-177,282,601	157	2,807x	✓ Covered
NSD1	chr5:177,283,767-177,283,948	182	3,171x	✓ Covered
NSD1	chr5:177,288,799-177,288,945	147	2,689x	✓ Covered
NSD1	chr5:177,291,934-177,292,178	245	3,836x	✓ Covered
NSD1	chr5:177,293,812-177,295,476	1,665	5,891x	✓ Covered
EGFR				
EGFR	chr7:55,173,901-55,174,063	163	2,952x	✓ Covered
EGFR	chr7:55,174,702-55,174,840	139	2,542x	✓ Covered
EGFR	chr7:55,181,273-55,181,498	226	3,670x	✓ Covered
EGFR	chr7:55,191,699-55,191,894	196	3,324x	✓ Covered
CDKN2A				
CDKN2A	chr9:21,968,212-21,968,262	51	1,127x	✓ Covered
CDKN2A	chr9:21,970,882-21,971,228	347	4,489x	✓ Covered
CDKN2A	chr9:21,974,658-21,974,846	189	6,485x	✓ Covered
CDKN2A	chr9:21,974,658-21,974,847	190	6,503x	✓ Covered
NOTCH1				
NOTCH1	chr9:136,496,054-136,497,578	1,525	5,847x	✓ Covered
NOTCH1	chr9:136,498,879-136,499,016	138	2,543x	✓ Covered
NOTCH1	chr9:136,499,092-136,499,279	188	4,412x	✓ Covered
NOTCH1	chr9:136,500,532-136,500,867	336	4,432x	✓ Covered
NOTCH1	chr9:136,501,728-136,501,933	206	3,471x	✓ Covered
NOTCH1	chr9:136,501,981-136,502,108	128	4,923x	✓ Covered
NOTCH1	chr9:136,502,252-136,502,508	257	4,254x	✓ Covered
NOTCH1	chr9:136,503,162-136,503,350	189	3,260x	✓ Covered
NOTCH1	chr9:136,504,653-136,505,124	472	4,906x	✓ Covered
NOTCH1	chr9:136,505,290-136,505,901	612	5,320x	✓ Covered
NOTCH1	chr9:136,506,507-136,506,659	153	2,759x	✓ Covered
NOTCH1	chr9:136,506,696-136,506,993	298	5,481x	✓ Covered
NOTCH1	chr9:136,507,285-136,507,457	173	3,112x	✓ Covered
NOTCH1	chr9:136,507,935-136,508,159	225	3,718x	✓ Covered
NOTCH1	chr9:136,508,212-136,508,405	194	5,001x	✓ Covered

FULL PANEL COVERAGE REPORT — Kafka, Franz J. (NGS-2026-052105)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
NOTCH1	chr9:136,508,850-136,509,091	242	3,793x	✓ Covered
NOTCH1	chr9:136,509,713-136,509,981	269	4,037x	✓ Covered
NOTCH1	chr9:136,510,633-136,510,825	193	3,324x	✓ Covered
NOTCH1	chr9:136,511,132-136,511,291	160	2,828x	✓ Covered
NOTCH1	chr9:136,513,001-136,513,154	154	2,743x	✓ Covered
NOTCH1	chr9:136,513,372-136,513,557	186	3,259x	✓ Covered
NOTCH1	chr9:136,514,490-136,514,722	233	3,782x	✓ Covered
NOTCH1	chr9:136,515,270-136,515,420	151	2,743x	✓ Covered
NOTCH1	chr9:136,515,463-136,515,736	274	5,297x	✓ Covered
NOTCH1	chr9:136,515,961-136,516,114	154	2,721x	✓ Covered
NOTCH1	chr9:136,517,252-136,517,405	154	2,798x	✓ Covered
NOTCH1	chr9:136,517,732-136,517,957	226	3,648x	✓ Covered
NOTCH1	chr9:136,518,117-136,518,312	196	3,553x	✓ Covered
NOTCH1	chr9:136,518,571-136,518,844	274	4,077x	✓ Covered
NOTCH1	chr9:136,519,423-136,519,585	163	2,912x	✓ Covered
NOTCH1	chr9:136,522,830-136,523,208	379	4,651x	✓ Covered
NOTCH1	chr9:136,523,697-136,523,999	303	4,324x	✓ Covered
NOTCH1	chr9:136,544,004-136,544,122	119	2,200x	✓ Covered
NOTCH1	chr9:136,545,706-136,545,806	101	1,932x	✓ Covered

PATIENT & SPECIMEN INFORMATION			
Patient name	Morrison, Toni E.	Accession	NGS-2026-052106
Date of birth	1955-10-09 (Age 70)	Specimen	Oral swab, left retromolar trigone
MRN	60890176	Collected	2026-05-19
Ordering physician	Dr. Wisdom, MD	Received	2026-05-19
Clinical indication	Oral lesion — left retromolar trigone	Reported	2026-05-20

RESULT: NO REPORTABLE VARIANTS DETECTED

INTERPRETATION

No pathogenic or likely pathogenic somatic variants were detected in the 21 genes covered by this assay at $\geq 0.2\%$ VAF with ≥ 2 consensus supporting reads. A negative result does not exclude malignancy. Variants below the reporting threshold, variants outside the 251 assay target regions, structural variants, and copy number alterations are not assessed by this assay. Clinical correlation is required; repeat testing or biopsy should be considered if clinical suspicion persists.

GENES WITH NO PATHOGENIC OR LIKELY PATHOGENIC VARIANTS DETECTED

AJUBA, CASP8, CDKN2A, CUL3, DPYD, EGFR, EP300, FAT1, FBXW7, HRAS, KEAP1, KMT2D, KRAS, NFE2L2, NOTCH1, NSD1, PIK3CA, PIK3R1, PTEN, TERT, TP53

Variants of uncertain significance (Tier III) are not reported.

METHOD SUMMARY

Cell-free and cellular DNA extracted from oral swab is subjected to hybridization capture using the IDT xGen Duplex UMI panel targeting 251 genomic regions (~77 kbp) across 21 cancer-relevant genes. Libraries incorporate 9-bp Unique Molecular Identifiers (UMIs) in a 9M1S+T read structure on both R1 and R2. Libraries are sequenced on Illumina NextSeq (2x150 bp). After alignment to GRCh38 (canonical chromosomes), reads are grouped by UMI using fgbio (adjacency strategy, ≤ 1 edit distance), and molecular consensus sequences are called requiring ≥ 2 reads per UMI family. Somatic variants are called on the consensus BAM with VarDict at minimum allele frequency 0.001. Variants $\geq 0.2\%$ consensus VAF with ≥ 2 consensus supporting reads are reported. UMI deduplication reduces sequencing noise by >800 -fold. Run-level negative controls are included on every sequencing run.

LIMITATIONS

This assay detects somatic single nucleotide variants and small indels within 251 target regions (~77 kbp) at VAF $\geq 0.2\%$ with validated sensitivity of 98.4% at $\geq 20,000\times$ raw input depth (~4,000 $\times$ consensus after UMI deduplication). Structural variants, copy number alterations, gene fusions, large indels, intronic variants outside defined capture regions, and variants in regions with low hybridization capture efficiency are not reliably assessed. A negative result does not exclude malignancy. Tumour heterogeneity, low tumour cell content, and specimen quality may affect variant allele fraction. Variants below the 0.2% VAF reporting threshold may be present but are not reported. At 20,000 $\times$ raw sequencing depth (~4,000 $\times$ mean consensus), approximately 0.3% of targets may fall below 1,000 $\times$ consensus depth; per-target coverage is provided in the laboratory technical report. Results should be interpreted in the context of clinical and pathological findings by a qualified clinician.

Reported by: Dr. Wisdom, MD PhD, FCAP

Report date: 2026-05-20

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

HOTSPOT SCREENING SUMMARY

Patient:

Morrison, Toni E.

Accession:

NGS-2026-052106

Hotspots listed below were explicitly interrogated in addition to all variants above 0.2% VAF with ≥2 consensus reads across all 251 targets. Coverage criteria (≥1,000× consensus depth) were met at all targets.

EGFR			Cov ✓
L858R	c.2573T>G	Not detected	
T790M	c.2369C>T	Not detected	
Exon 19 del	c.2235_2249del	Not detected	
KRAS			Cov ✓
G12D	c.35G>T	Not detected	
G12V	c.35G>A	Not detected	
G12C	c.34G>T	Not detected	
G13D	c.38G>A	Not detected	
PIK3CA			Cov ✓
E542K	c.1624G>A	Not detected	
E545K	c.1633G>A	Not detected	
H1047R	c.3140A>G	Not detected	
TERT			Cov ✓
C228T	c.-124C>T	Not detected	
C250T	c.-146C>T	Not detected	
HRAS			Cov ✓
G12V	c.35G>T	Not detected	
G12D	c.35G>A	Not detected	
Q61L	c.182A>T	Not detected	
Q61R	c.182A>G	Not detected	
NOTCH1			Cov ✓
P2415L	c.7244C>T	Not detected	
R1699W	c.5095C>T	Not detected	
TP53			Cov ✓
R175H	c.524G>A	Not detected	
R248Q	c.743G>A	Not detected	
R248W	c.742C>T	Not detected	
R273H	c.818G>A	Not detected	
R282W	c.844C>T	Not detected	

REMAINING PANEL GENES — SEQUENCE-LEVEL DETECTION, NO PREDEFINED HOTSPOT LIST

AJUBA · CASP8 · CDKN2A · CUL3 · DPYD · EP300 · FAT1 · FBXW7 · KEAP1 · KMT2D · NFE2L2 · NSD1 · PIK3R1 · PTEN

All variants above 0.2% VAF with ≥2 consensus reads are reported for these genes. No variants meeting reporting criteria were detected. All targets in these genes met minimum coverage criteria (≥1,000× consensus depth).

Reported by: Dr. Wisdom, MD PhD, FCAP

Report date: 2026-05-20

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

FULL PANEL COVERAGE REPORT — Morrison, Toni E. (NGS-2026-052106)

251 targets, 77,417 bp total. Mean consensus depth: 3,665x. Targets below 1,000x consensus: 0 / 251 (0%). Depths are from the UMI-deduplicated consensus BAM (min-reads ≥ 2, 20,000x raw input). Positions reported 1-based.

Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
DPYD				
DPYD	chr1:97,082,291-97,082,491	201	3,447x	✓ Covered
DPYD	chr1:97,449,958-97,450,158	201	3,492x	✓ Covered
DPYD	chr1:97,515,687-97,515,887	201	3,452x	✓ Covered
DPYD	chr1:97,573,763-97,573,963	201	3,469x	✓ Covered
PTEN				
PTEN	chr10:87,864,450-87,864,568	119	2,266x	✓ Covered
PTEN	chr10:87,894,005-87,894,129	125	2,315x	✓ Covered
PTEN	chr10:87,925,493-87,925,577	85	1,735x	✓ Covered
PTEN	chr10:87,931,026-87,931,109	84	1,637x	✓ Covered
PTEN	chr10:87,932,993-87,933,271	279	4,109x	✓ Covered
PTEN	chr10:87,952,098-87,952,279	182	3,186x	✓ Covered
PTEN	chr10:87,957,833-87,958,039	207	3,517x	✓ Covered
PTEN	chr10:87,960,874-87,961,138	265	4,049x	✓ Covered
PTEN	chr10:87,965,267-87,965,489	223	3,488x	✓ Covered
HRAS				
HRAS	chr11:532,619-532,775	157	2,809x	✓ Covered
HRAS	chr11:533,433-533,632	200	3,417x	✓ Covered
HRAS	chr11:533,746-533,964	219	4,200x	✓ Covered
HRAS	chr11:534,192-534,342	151	2,709x	✓ Covered
KRAS				
KRAS	chr12:25,209,778-25,209,931	154	2,740x	✓ Covered
KRAS	chr12:25,225,594-25,225,793	200	3,379x	✓ Covered
KRAS	chr12:25,227,214-25,227,432	219	3,583x	✓ Covered
KRAS	chr12:25,245,254-25,245,404	151	2,699x	✓ Covered
KMT2D				
KMT2D	chr12:49,021,763-49,021,892	130	2,432x	✓ Covered
KMT2D	chr12:49,022,023-49,022,171	149	3,251x	✓ Covered
KMT2D	chr12:49,022,260-49,022,373	114	3,798x	✓ Covered
KMT2D	chr12:49,022,570-49,022,895	326	4,424x	✓ Covered
KMT2D	chr12:49,024,558-49,024,728	171	3,055x	✓ Covered
KMT2D	chr12:49,024,790-49,024,966	177	4,689x	✓ Covered
KMT2D	chr12:49,026,162-49,027,342	1,181	5,736x	✓ Covered
KMT2D	chr12:49,027,783-49,027,950	168	2,979x	✓ Covered
KMT2D	chr12:49,027,989-49,028,161	173	5,128x	✓ Covered
KMT2D	chr12:49,028,808-49,028,978	171	3,017x	✓ Covered
KMT2D	chr12:49,029,041-49,029,256	216	4,894x	✓ Covered
KMT2D	chr12:49,029,381-49,029,496	116	3,144x	✓ Covered
KMT2D	chr12:49,030,260-49,030,459	200	3,398x	✓ Covered
KMT2D	chr12:49,030,581-49,030,788	208	4,040x	✓ Covered
KMT2D	chr12:49,030,873-49,031,053	181	4,349x	✓ Covered
KMT2D	chr12:49,031,155-49,033,984	2,830	6,140x	✓ Covered
KMT2D	chr12:49,034,047-49,034,319	273	5,350x	✓ Covered
KMT2D	chr12:49,034,390-49,034,496	107	4,455x	✓ Covered
KMT2D	chr12:49,034,562-49,034,686	125	4,131x	✓ Covered
KMT2D	chr12:49,034,792-49,034,955	164	3,734x	✓ Covered
KMT2D	chr12:49,037,105-49,039,009	1,905	5,978x	✓ Covered
KMT2D	chr12:49,039,202-49,039,378	177	3,187x	✓ Covered
KMT2D	chr12:49,039,415-49,039,637	223	5,413x	✓ Covered

FULL PANEL COVERAGE REPORT — Morrison, Toni E. (NGS-2026-052106)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
KMT2D	chr12:49,039,704–49,041,555	1,852	6,108×	✓ Covered
KMT2D	chr12:49,041,635–49,041,725	91	4,392×	✓ Covered
KMT2D	chr12:49,041,897–49,042,010	114	2,328×	✓ Covered
KMT2D	chr12:49,042,069–49,042,350	282	5,076×	✓ Covered
KMT2D	chr12:49,042,541–49,042,665	125	2,378×	✓ Covered
KMT2D	chr12:49,042,721–49,042,898	178	4,621×	✓ Covered
KMT2D	chr12:49,043,056–49,043,206	151	3,074×	✓ Covered
KMT2D	chr12:49,043,343–49,043,448	106	2,727×	✓ Covered
KMT2D	chr12:49,043,615–49,043,802	188	3,399×	✓ Covered
KMT2D	chr12:49,043,848–49,044,018	171	5,094×	✓ Covered
KMT2D	chr12:49,044,180–49,044,324	145	2,914×	✓ Covered
KMT2D	chr12:49,044,383–49,044,542	160	4,601×	✓ Covered
KMT2D	chr12:49,044,724–49,044,985	262	4,031×	✓ Covered
KMT2D	chr12:49,045,900–49,045,987	88	1,673×	✓ Covered
KMT2D	chr12:49,046,045–49,046,194	150	4,206×	✓ Covered
KMT2D	chr12:49,046,240–49,046,444	205	5,053×	✓ Covered
KMT2D	chr12:49,046,589–49,046,810	222	3,905×	✓ Covered
KMT2D	chr12:49,047,945–49,048,089	145	2,582×	✓ Covered
KMT2D	chr12:49,048,639–49,048,789	151	2,714×	✓ Covered
KMT2D	chr12:49,049,085–49,049,238	154	2,770×	✓ Covered
KMT2D	chr12:49,049,662–49,050,810	1,149	5,674×	✓ Covered
KMT2D	chr12:49,050,866–49,052,444	1,579	6,129×	✓ Covered
KMT2D	chr12:49,052,544–49,052,729	186	4,423×	✓ Covered
KMT2D	chr12:49,052,895–49,053,092	198	3,575×	✓ Covered
KMT2D	chr12:49,053,187–49,053,341	155	3,994×	✓ Covered
KMT2D	chr12:49,053,456–49,053,661	206	4,052×	✓ Covered
KMT2D	chr12:49,053,958–49,054,160	203	3,441×	✓ Covered
KMT2D	chr12:49,054,287–49,054,436	150	3,407×	✓ Covered
KMT2D	chr12:49,054,508–49,054,771	264	4,969×	✓ Covered
KMT2D	chr12:49,054,880–49,055,046	167	3,825×	✓ Covered
KMT2D	chr12:49,055,256–49,055,344	89	1,797×	✓ Covered
AJUBA				
AJUBA	chr14:22,973,426–22,973,588	163	2,853×	✓ Covered
AJUBA	chr14:22,974,027–22,974,135	109	2,052×	✓ Covered
AJUBA	chr14:22,974,819–22,974,910	92	1,796×	✓ Covered
AJUBA	chr14:22,974,954–22,975,124	171	4,627×	✓ Covered
AJUBA	chr14:22,976,436–22,976,538	103	1,907×	✓ Covered
AJUBA	chr14:22,976,625–22,976,732	108	3,525×	✓ Covered
AJUBA	chr14:22,978,324–22,978,465	142	2,602×	✓ Covered
AJUBA	chr14:22,981,241–22,982,286	1,046	5,685×	✓ Covered
TP53				
TP53	chr17:7,669,592–7,669,710	119	2,256×	✓ Covered
TP53	chr17:7,670,589–7,670,735	147	2,675×	✓ Covered
TP53	chr17:7,673,515–7,673,628	114	2,162×	✓ Covered
TP53	chr17:7,673,681–7,673,857	177	4,731×	✓ Covered
TP53	chr17:7,674,161–7,674,310	150	2,768×	✓ Covered
TP53	chr17:7,674,839–7,674,991	153	2,787×	✓ Covered
TP53	chr17:7,675,033–7,675,256	224	5,175×	✓ Covered
TP53	chr17:7,675,974–7,676,292	319	4,362×	✓ Covered
TP53	chr17:7,676,362–7,676,423	62	4,264×	✓ Covered
TP53	chr17:7,676,501–7,676,614	114	3,454×	✓ Covered

FULL PANEL COVERAGE REPORT — Morrison, Toni E. (NGS-2026-052106)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
KEAP1				
KEAP1	chr19:10,486,635–10,486,838	204	3,492×	✓ Covered
KEAP1	chr19:10,489,172–10,489,388	217	3,618×	✓ Covered
KEAP1	chr19:10,489,628–10,489,873	246	3,821×	✓ Covered
KEAP1	chr19:10,491,557–10,492,282	726	5,412×	✓ Covered
KEAP1	chr19:10,499,375–10,500,053	679	5,348×	✓ Covered
NFE2L2				
NFE2L2	chr2:177,233,985–177,234,291	307	4,321×	✓ Covered
CASP8				
CASP8	chr2:201,266,467–201,266,811	345	4,519×	✓ Covered
CASP8	chr2:201,271,496–201,271,641	146	2,672×	✓ Covered
CASP8	chr2:201,272,618–201,272,796	179	3,129×	✓ Covered
CASP8	chr2:201,272,878–201,272,962	85	3,868×	✓ Covered
CASP8	chr2:201,274,869–201,274,973	105	1,970×	✓ Covered
CASP8	chr2:201,276,807–201,276,988	182	3,160×	✓ Covered
CASP8	chr2:201,284,796–201,285,337	542	5,070×	✓ Covered
CASP8	chr2:201,286,439–201,286,611	173	3,024×	✓ Covered
CUL3				
CUL3	chr2:224,474,228–224,474,396	169	3,019×	✓ Covered
CUL3	chr2:224,478,180–224,478,365	186	3,209×	✓ Covered
CUL3	chr2:224,481,872–224,482,098	227	3,685×	✓ Covered
CUL3	chr2:224,495,812–224,495,986	175	3,148×	✓ Covered
CUL3	chr2:224,497,733–224,497,869	137	2,526×	✓ Covered
CUL3	chr2:224,500,343–224,500,507	165	2,905×	✓ Covered
CUL3	chr2:224,502,945–224,503,092	148	2,656×	✓ Covered
CUL3	chr2:224,503,632–224,503,842	211	3,515×	✓ Covered
CUL3	chr2:224,505,936–224,506,152	217	3,660×	✓ Covered
CUL3	chr2:224,506,838–224,507,023	186	3,208×	✓ Covered
CUL3	chr2:224,511,334–224,511,602	269	4,074×	✓ Covered
CUL3	chr2:224,513,504–224,513,658	155	2,779×	✓ Covered
CUL3	chr2:224,514,592–224,514,792	201	3,422×	✓ Covered
CUL3	chr2:224,535,508–224,535,661	154	2,740×	✓ Covered
CUL3	chr2:224,557,639–224,557,876	238	3,742×	✓ Covered
CUL3	chr2:224,584,924–224,585,029	106	2,075×	✓ Covered
EP300				
EP300	chr22:41,092,985–41,093,118	134	2,472×	✓ Covered
EP300	chr22:41,117,167–41,117,841	675	5,340×	✓ Covered
EP300	chr22:41,125,844–41,126,060	217	3,561×	✓ Covered
EP300	chr22:41,127,467–41,127,768	302	4,193×	✓ Covered
EP300	chr22:41,129,870–41,130,023	154	2,741×	✓ Covered
EP300	chr22:41,131,368–41,131,653	286	4,139×	✓ Covered
EP300	chr22:41,135,793–41,135,926	134	2,478×	✓ Covered
EP300	chr22:41,137,633–41,137,810	178	3,079×	✓ Covered
EP300	chr22:41,140,120–41,140,277	158	2,831×	✓ Covered
EP300	chr22:41,141,028–41,141,242	215	3,602×	✓ Covered
EP300	chr22:41,146,719–41,146,836	118	2,254×	✓ Covered
EP300	chr22:41,147,817–41,147,966	150	2,752×	✓ Covered
EP300	chr22:41,149,018–41,149,195	178	3,089×	✓ Covered
EP300	chr22:41,149,741–41,150,218	478	4,910×	✓ Covered
EP300	chr22:41,151,813–41,152,032	220	3,637×	✓ Covered
EP300	chr22:41,152,186–41,152,370	185	3,488×	✓ Covered

FULL PANEL COVERAGE REPORT — Morrison, Toni E. (NGS-2026-052106)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
EP300	chr22:41,154,975-41,155,133	159	2,864×	✓ Covered
EP300	chr22:41,157,149-41,157,428	280	4,101×	✓ Covered
EP300	chr22:41,158,392-41,158,520	129	2,380×	✓ Covered
EP300	chr22:41,160,622-41,160,742	121	2,254×	✓ Covered
EP300	chr22:41,162,703-41,162,799	97	1,911×	✓ Covered
EP300	chr22:41,164,033-41,164,150	118	2,197×	✓ Covered
EP300	chr22:41,166,579-41,166,686	108	2,060×	✓ Covered
EP300	chr22:41,168,429-41,168,619	191	3,321×	✓ Covered
EP300	chr22:41,168,701-41,168,887	187	4,397×	✓ Covered
EP300	chr22:41,169,483-41,169,636	154	2,788×	✓ Covered
EP300	chr22:41,170,386-41,170,591	206	3,482×	✓ Covered
EP300	chr22:41,172,479-41,172,683	205	3,530×	✓ Covered
EP300	chr22:41,173,603-41,173,804	202	3,419×	✓ Covered
EP300	chr22:41,176,227-41,176,548	322	4,352×	✓ Covered
EP300	chr22:41,176,753-41,178,973	2,221	5,999×	✓ Covered
PIK3CA				
PIK3CA	chr3:179,218,190-179,218,354	165	2,906×	✓ Covered
PIK3CA	chr3:179,234,074-179,234,381	308	4,335×	✓ Covered
FBXW7				
FBXW7	chr4:152,325,986-152,326,251	266	3,985×	✓ Covered
FAT1				
FAT1	chr4:186,588,575-186,589,240	666	5,332×	✓ Covered
FAT1	chr4:186,595,669-186,595,846	178	3,147×	✓ Covered
FAT1	chr4:186,596,520-186,597,191	672	5,349×	✓ Covered
FAT1	chr4:186,597,662-186,597,812	151	2,700×	✓ Covered
FAT1	chr4:186,597,952-186,598,145	194	3,665×	✓ Covered
FAT1	chr4:186,599,878-186,600,380	503	5,026×	✓ Covered
FAT1	chr4:186,601,249-186,601,446	198	3,399×	✓ Covered
FAT1	chr4:186,602,883-186,603,054	172	3,057×	✓ Covered
FAT1	chr4:186,603,156-186,603,997	842	5,691×	✓ Covered
FAT1	chr4:186,604,357-186,604,594	238	3,833×	✓ Covered
FAT1	chr4:186,606,050-186,606,233	184	3,186×	✓ Covered
FAT1	chr4:186,609,163-186,609,340	178	3,106×	✓ Covered
FAT1	chr4:186,609,781-186,610,035	255	3,909×	✓ Covered
FAT1	chr4:186,611,366-186,611,795	430	4,793×	✓ Covered
FAT1	chr4:186,613,089-186,613,362	274	4,074×	✓ Covered
FAT1	chr4:186,614,171-186,614,364	194	3,334×	✓ Covered
FAT1	chr4:186,616,985-186,617,221	237	3,711×	✓ Covered
FAT1	chr4:186,617,688-186,621,795	4,108	6,168×	✓ Covered
FAT1	chr4:186,628,134-186,628,384	251	3,895×	✓ Covered
FAT1	chr4:186,628,468-186,628,783	316	5,071×	✓ Covered
FAT1	chr4:186,633,664-186,633,843	180	3,147×	✓ Covered
FAT1	chr4:186,636,005-186,636,255	251	3,882×	✓ Covered
FAT1	chr4:186,636,565-186,636,934	370	4,631×	✓ Covered
FAT1	chr4:186,639,702-186,639,803	102	1,970×	✓ Covered
FAT1	chr4:186,663,279-186,663,633	355	4,567×	✓ Covered
FAT1	chr4:186,706,543-186,709,847	3,305	6,115×	✓ Covered
TERT				
TERT	chr5:1,295,050-1,295,250	201	3,405×	✓ Covered
PIK3R1				
PIK3R1	chr5:68,294,516-68,294,698	183	3,173×	✓ Covered

FULL PANEL COVERAGE REPORT — Morrison, Toni E. (NGS-2026-052106)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
PIK3R1	chr5:68,295,128-68,295,344	217	3,678x	✓ Covered
PIK3R1	chr5:68,295,400-68,295,508	109	4,568x	✓ Covered
NSD1				
NSD1	chr5:177,135,084-177,136,050	967	5,610x	✓ Covered
NSD1	chr5:177,191,864-177,192,039	176	3,087x	✓ Covered
NSD1	chr5:177,204,100-177,204,312	213	3,577x	✓ Covered
NSD1	chr5:177,209,616-177,212,215	2,600	6,062x	✓ Covered
NSD1	chr5:177,235,801-177,235,965	165	2,957x	✓ Covered
NSD1	chr5:177,238,217-177,238,527	311	4,266x	✓ Covered
NSD1	chr5:177,239,736-177,239,885	150	2,656x	✓ Covered
NSD1	chr5:177,244,175-177,244,290	116	2,151x	✓ Covered
NSD1	chr5:177,246,658-177,246,816	159	2,791x	✓ Covered
NSD1	chr5:177,248,161-177,248,344	184	3,223x	✓ Covered
NSD1	chr5:177,251,710-177,251,873	164	2,915x	✓ Covered
NSD1	chr5:177,256,931-177,257,171	241	3,852x	✓ Covered
NSD1	chr5:177,259,969-177,260,188	220	3,622x	✓ Covered
NSD1	chr5:177,267,542-177,267,738	197	3,379x	✓ Covered
NSD1	chr5:177,269,582-177,269,827	246	3,834x	✓ Covered
NSD1	chr5:177,273,652-177,273,804	153	2,695x	✓ Covered
NSD1	chr5:177,280,545-177,280,854	310	4,311x	✓ Covered
NSD1	chr5:177,282,445-177,282,601	157	2,852x	✓ Covered
NSD1	chr5:177,283,767-177,283,948	182	3,196x	✓ Covered
NSD1	chr5:177,288,799-177,288,945	147	2,636x	✓ Covered
NSD1	chr5:177,291,934-177,292,178	245	3,907x	✓ Covered
NSD1	chr5:177,293,812-177,295,476	1,665	5,905x	✓ Covered
EGFR				
EGFR	chr7:55,173,901-55,174,063	163	2,949x	✓ Covered
EGFR	chr7:55,174,702-55,174,840	139	2,511x	✓ Covered
EGFR	chr7:55,181,273-55,181,498	226	3,710x	✓ Covered
EGFR	chr7:55,191,699-55,191,894	196	3,333x	✓ Covered
CDKN2A				
CDKN2A	chr9:21,968,212-21,968,262	51	1,105x	✓ Covered
CDKN2A	chr9:21,970,882-21,971,228	347	4,479x	✓ Covered
CDKN2A	chr9:21,974,658-21,974,846	189	6,606x	✓ Covered
CDKN2A	chr9:21,974,658-21,974,847	190	6,624x	✓ Covered
NOTCH1				
NOTCH1	chr9:136,496,054-136,497,578	1,525	5,855x	✓ Covered
NOTCH1	chr9:136,498,879-136,499,016	138	2,546x	✓ Covered
NOTCH1	chr9:136,499,092-136,499,279	188	4,411x	✓ Covered
NOTCH1	chr9:136,500,532-136,500,867	336	4,444x	✓ Covered
NOTCH1	chr9:136,501,728-136,501,933	206	3,399x	✓ Covered
NOTCH1	chr9:136,501,981-136,502,108	128	4,868x	✓ Covered
NOTCH1	chr9:136,502,252-136,502,508	257	4,253x	✓ Covered
NOTCH1	chr9:136,503,162-136,503,350	189	3,266x	✓ Covered
NOTCH1	chr9:136,504,653-136,505,124	472	4,940x	✓ Covered
NOTCH1	chr9:136,505,290-136,505,901	612	5,252x	✓ Covered
NOTCH1	chr9:136,506,507-136,506,659	153	2,747x	✓ Covered
NOTCH1	chr9:136,506,696-136,506,993	298	5,512x	✓ Covered
NOTCH1	chr9:136,507,285-136,507,457	173	3,059x	✓ Covered
NOTCH1	chr9:136,507,935-136,508,159	225	3,679x	✓ Covered
NOTCH1	chr9:136,508,212-136,508,405	194	5,011x	✓ Covered

FULL PANEL COVERAGE REPORT — Morrison, Toni E. (NGS-2026-052106)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
NOTCH1	chr9:136,508,850-136,509,091	242	3,796×	✓ Covered
NOTCH1	chr9:136,509,713-136,509,981	269	4,016×	✓ Covered
NOTCH1	chr9:136,510,633-136,510,825	193	3,294×	✓ Covered
NOTCH1	chr9:136,511,132-136,511,291	160	2,851×	✓ Covered
NOTCH1	chr9:136,513,001-136,513,154	154	2,797×	✓ Covered
NOTCH1	chr9:136,513,372-136,513,557	186	3,268×	✓ Covered
NOTCH1	chr9:136,514,490-136,514,722	233	3,746×	✓ Covered
NOTCH1	chr9:136,515,270-136,515,420	151	2,738×	✓ Covered
NOTCH1	chr9:136,515,463-136,515,736	274	5,348×	✓ Covered
NOTCH1	chr9:136,515,961-136,516,114	154	2,820×	✓ Covered
NOTCH1	chr9:136,517,252-136,517,405	154	2,807×	✓ Covered
NOTCH1	chr9:136,517,732-136,517,957	226	3,657×	✓ Covered
NOTCH1	chr9:136,518,117-136,518,312	196	3,606×	✓ Covered
NOTCH1	chr9:136,518,571-136,518,844	274	4,068×	✓ Covered
NOTCH1	chr9:136,519,423-136,519,585	163	2,918×	✓ Covered
NOTCH1	chr9:136,522,830-136,523,208	379	4,590×	✓ Covered
NOTCH1	chr9:136,523,697-136,523,999	303	4,258×	✓ Covered
NOTCH1	chr9:136,544,004-136,544,122	119	2,225×	✓ Covered
NOTCH1	chr9:136,545,706-136,545,806	101	1,941×	✓ Covered

PATIENT & SPECIMEN INFORMATION

Patient name	Austen, Jane C.	Accession	NGS-2026-052104
Date of birth	1967-12-04 (Age 58)	Specimen	Oral swab, soft palate
MRN	40678954	Collected	2026-05-19
Ordering physician	Dr. Wisdom, MD	Received	2026-05-19
Clinical indication	Oral lesion — soft palate	Reported	2026-05-20

RESULT: VARIANTS DETECTED

DETECTED VARIANTS

Gene	HGVS (coding)	Protein change	VAF	Depth	Tier	Classification
EGFR	NM_005228.5:c.2573T>G	p.Leu858Arg	0.57%	3,351x	I	Pathogenic
TP53	NM_000546.6:c.524G>A	p.Arg175His	0.30%	5,229x	I	Pathogenic

VARIANT ANNOTATION

EGFR p.Leu858Arg | Pathogenic (Tier I)

EGFR p.Leu858Arg (L858R) is a well-characterised activating mutation in exon 21 of the EGFR kinase domain. It disrupts the autoinhibitory conformation, leading to constitutive EGFR kinase activation and downstream RAS/MAPK and PI3K/AKT signalling. While canonical in non-small cell lung adenocarcinoma (where it predicts response to EGFR TKIs such as osimertinib), EGFR L858R has been reported in 1–3% of OSCC. Cetuximab (anti-EGFR monoclonal antibody) is FDA-approved in HNSCC and may have enhanced activity in the setting of EGFR activating mutations. Multidisciplinary tumour board review for consideration of EGFR-directed therapy or clinical trial enrolment is recommended. Co-occurring TP53 R175H (see below) may affect response to EGFR-directed therapy and warrants combined consideration.

TP53 p.Arg175His | Pathogenic (Tier I)

TP53 p.Arg175His — see interpretation above (Dickens, Charles J., NGS-2026-052103). Co-occurrence of EGFR L858R and TP53 R175H has been associated with resistance to EGFR-directed therapies in other tumour types; multidisciplinary tumour board review is recommended.

GENES WITH NO PATHOGENIC OR LIKELY PATHOGENIC VARIANTS DETECTED

AJUBA, CASP8, CDKN2A, CUL3, DPYD, EP300, FAT1, FBXW7, HRAS, KEAP1, KMT2D, KRAS, NFE2L2, NOTCH1, NSD1, PIK3CA, PIK3R1, PTEN, TERT

Variants of uncertain significance (Tier III) are not reported.

METHOD SUMMARY

Cell-free and cellular DNA extracted from oral swab is subjected to hybridization capture using the IDT xGen Duplex UMI panel targeting 251 genomic regions (~77 kbp) across 21 cancer-relevant genes. Libraries incorporate 9-bp Unique Molecular Identifiers (UMIs) in a 9M1S+T read structure on both R1 and R2. Libraries are sequenced on Illumina NextSeq (2x150 bp). After alignment to GRCh38 (canonical chromosomes), reads are grouped by UMI using fgbio (adjacency strategy, ≤1 edit distance), and molecular consensus sequences are called requiring ≥2 reads per UMI family. Somatic variants are called on the consensus BAM with VarDict at minimum allele frequency 0.001. Variants ≥0.2% consensus VAF with ≥2 consensus supporting reads are reported. UMI deduplication reduces sequencing noise by >800-fold. Run-level negative controls are included on every sequencing run.

LIMITATIONS

This assay detects somatic single nucleotide variants and small indels within 251 target regions (~77 kbp) at VAF ≥0.2% with validated sensitivity of 98.4% at ≥20,000x raw input depth (~4,000x consensus after UMI deduplication). Structural variants, copy number alterations, gene fusions, large indels, intronic variants outside defined capture regions, and variants in regions with low hybridization capture efficiency are not reliably assessed. A negative result does not exclude malignancy. Tumour heterogeneity, low tumour cell content, and specimen quality may affect variant allele fraction. Variants below the 0.2% VAF reporting threshold may be present but are not reported. At 20,000x raw sequencing depth (~4,000x mean consensus), approximately 0.3% of targets may fall below 1,000x consensus depth; per-target coverage is provided in the laboratory technical report. Results should be interpreted in the context of clinical and pathological findings by a qualified clinician.

Reported by: Dr. Wisdom, MD PhD, FCAP

Report date: 2026-05-20

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

HOTSPOT SCREENING SUMMARY

Patient:

Austen, Jane C.

Accession:

NGS-2026-052104

Hotspots listed below were explicitly interrogated in addition to all variants above 0.2% VAF with ≥2 consensus reads across all 251 targets. Coverage criteria (≥1,000× consensus depth) were met at all targets.

EGFR			Cov ✓
L858R	c.2573T>G	● DETECTED	
T790M	c.2369C>T	Not detected	
Exon 19 del	c.2235_2249del	Not detected	

KRAS			Cov ✓
G12D	c.35G>T	Not detected	
G12V	c.35G>A	Not detected	
G12C	c.34G>T	Not detected	
G13D	c.38G>A	Not detected	

PIK3CA			Cov ✓
E542K	c.1624G>A	Not detected	
E545K	c.1633G>A	Not detected	
H1047R	c.3140A>G	Not detected	

TERT			Cov ✓
C228T	c.-124C>T	Not detected	
C250T	c.-146C>T	Not detected	

HRAS			Cov ✓
G12V	c.35G>T	Not detected	
G12D	c.35G>A	Not detected	
Q61L	c.182A>T	Not detected	
Q61R	c.182A>G	Not detected	

NOTCH1			Cov ✓
P2415L	c.7244C>T	Not detected	
R1699W	c.5095C>T	Not detected	

TP53			Cov ✓
R175H	c.524G>A	● DETECTED	
R248Q	c.743G>A	Not detected	
R248W	c.742C>T	Not detected	
R273H	c.818G>A	Not detected	
R282W	c.844C>T	Not detected	

REMAINING PANEL GENES — SEQUENCE-LEVEL DETECTION, NO PREDEFINED HOTSPOT LIST

AJUBA · CASP8 · CDKN2A · CUL3 · DPYD · EP300 · FAT1 · FBXW7 · KEAP1 · KMT2D · NFE2L2 · NSD1 · PIK3R1 · PTEN

All variants above 0.2% VAF with ≥2 consensus reads are reported for these genes. No variants meeting reporting criteria were detected. All targets in these genes met minimum coverage criteria (≥1,000× consensus depth).

Reported by: Dr. Wisdom, MD PhD, FCAP

Report date: 2026-05-20

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

FULL PANEL COVERAGE REPORT — Austen, Jane C. (NGS-2026-052104)

251 targets, 77,417 bp total. Mean consensus depth: 3,667x. Targets below 1,000x consensus: 0 / 251 (0%). Depths are from the UMI-deduplicated consensus BAM (min-reads ≥ 2, 20,000x raw input). Positions reported 1-based.

Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
DPYD				
DPYD	chr1:97,082,291-97,082,491	201	3,382x	✓ Covered
DPYD	chr1:97,449,958-97,450,158	201	3,432x	✓ Covered
DPYD	chr1:97,515,687-97,515,887	201	3,382x	✓ Covered
DPYD	chr1:97,573,763-97,573,963	201	3,349x	✓ Covered
PTEN				
PTEN	chr10:87,864,450-87,864,568	119	2,203x	✓ Covered
PTEN	chr10:87,894,005-87,894,129	125	2,353x	✓ Covered
PTEN	chr10:87,925,493-87,925,577	85	1,712x	✓ Covered
PTEN	chr10:87,931,026-87,931,109	84	1,674x	✓ Covered
PTEN	chr10:87,932,993-87,933,271	279	4,123x	✓ Covered
PTEN	chr10:87,952,098-87,952,279	182	3,122x	✓ Covered
PTEN	chr10:87,957,833-87,958,039	207	3,511x	✓ Covered
PTEN	chr10:87,960,874-87,961,138	265	4,003x	✓ Covered
PTEN	chr10:87,965,267-87,965,489	223	3,470x	✓ Covered
HRAS				
HRAS	chr11:532,619-532,775	157	2,823x	✓ Covered
HRAS	chr11:533,433-533,632	200	3,375x	✓ Covered
HRAS	chr11:533,746-533,964	219	4,167x	✓ Covered
HRAS	chr11:534,192-534,342	151	2,706x	✓ Covered
KRAS				
KRAS	chr12:25,209,778-25,209,931	154	2,777x	✓ Covered
KRAS	chr12:25,225,594-25,225,793	200	3,406x	✓ Covered
KRAS	chr12:25,227,214-25,227,432	219	3,578x	✓ Covered
KRAS	chr12:25,245,254-25,245,404	151	2,704x	✓ Covered
KMT2D				
KMT2D	chr12:49,021,763-49,021,892	130	2,405x	✓ Covered
KMT2D	chr12:49,022,023-49,022,171	149	3,322x	✓ Covered
KMT2D	chr12:49,022,260-49,022,373	114	3,740x	✓ Covered
KMT2D	chr12:49,022,570-49,022,895	326	4,435x	✓ Covered
KMT2D	chr12:49,024,558-49,024,728	171	3,066x	✓ Covered
KMT2D	chr12:49,024,790-49,024,966	177	4,734x	✓ Covered
KMT2D	chr12:49,026,162-49,027,342	1,181	5,767x	✓ Covered
KMT2D	chr12:49,027,783-49,027,950	168	3,042x	✓ Covered
KMT2D	chr12:49,027,989-49,028,161	173	5,106x	✓ Covered
KMT2D	chr12:49,028,808-49,028,978	171	3,052x	✓ Covered
KMT2D	chr12:49,029,041-49,029,256	216	4,857x	✓ Covered
KMT2D	chr12:49,029,381-49,029,496	116	3,179x	✓ Covered
KMT2D	chr12:49,030,260-49,030,459	200	3,436x	✓ Covered
KMT2D	chr12:49,030,581-49,030,788	208	4,070x	✓ Covered
KMT2D	chr12:49,030,873-49,031,053	181	4,353x	✓ Covered
KMT2D	chr12:49,031,155-49,033,984	2,830	6,146x	✓ Covered
KMT2D	chr12:49,034,047-49,034,319	273	5,393x	✓ Covered
KMT2D	chr12:49,034,390-49,034,496	107	4,431x	✓ Covered
KMT2D	chr12:49,034,562-49,034,686	125	4,125x	✓ Covered
KMT2D	chr12:49,034,792-49,034,955	164	3,771x	✓ Covered
KMT2D	chr12:49,037,105-49,039,009	1,905	5,957x	✓ Covered
KMT2D	chr12:49,039,202-49,039,378	177	3,214x	✓ Covered
KMT2D	chr12:49,039,415-49,039,637	223	5,392x	✓ Covered

FULL PANEL COVERAGE REPORT — Austen, Jane C. (NGS-2026-052104)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
KMT2D	chr12:49,039,704–49,041,555	1,852	6,093×	✓ Covered
KMT2D	chr12:49,041,635–49,041,725	91	4,474×	✓ Covered
KMT2D	chr12:49,041,897–49,042,010	114	2,370×	✓ Covered
KMT2D	chr12:49,042,069–49,042,350	282	4,999×	✓ Covered
KMT2D	chr12:49,042,541–49,042,665	125	2,421×	✓ Covered
KMT2D	chr12:49,042,721–49,042,898	178	4,668×	✓ Covered
KMT2D	chr12:49,043,056–49,043,206	151	3,015×	✓ Covered
KMT2D	chr12:49,043,343–49,043,448	106	2,748×	✓ Covered
KMT2D	chr12:49,043,615–49,043,802	188	3,439×	✓ Covered
KMT2D	chr12:49,043,848–49,044,018	171	5,096×	✓ Covered
KMT2D	chr12:49,044,180–49,044,324	145	2,911×	✓ Covered
KMT2D	chr12:49,044,383–49,044,542	160	4,590×	✓ Covered
KMT2D	chr12:49,044,724–49,044,985	262	4,028×	✓ Covered
KMT2D	chr12:49,045,900–49,045,987	88	1,750×	✓ Covered
KMT2D	chr12:49,046,045–49,046,194	150	4,294×	✓ Covered
KMT2D	chr12:49,046,240–49,046,444	205	4,988×	✓ Covered
KMT2D	chr12:49,046,589–49,046,810	222	3,971×	✓ Covered
KMT2D	chr12:49,047,945–49,048,089	145	2,649×	✓ Covered
KMT2D	chr12:49,048,639–49,048,789	151	2,761×	✓ Covered
KMT2D	chr12:49,049,085–49,049,238	154	2,726×	✓ Covered
KMT2D	chr12:49,049,662–49,050,810	1,149	5,726×	✓ Covered
KMT2D	chr12:49,050,866–49,052,444	1,579	6,150×	✓ Covered
KMT2D	chr12:49,052,544–49,052,729	186	4,294×	✓ Covered
KMT2D	chr12:49,052,895–49,053,092	198	3,607×	✓ Covered
KMT2D	chr12:49,053,187–49,053,341	155	3,913×	✓ Covered
KMT2D	chr12:49,053,456–49,053,661	206	4,083×	✓ Covered
KMT2D	chr12:49,053,958–49,054,160	203	3,420×	✓ Covered
KMT2D	chr12:49,054,287–49,054,436	150	3,419×	✓ Covered
KMT2D	chr12:49,054,508–49,054,771	264	4,922×	✓ Covered
KMT2D	chr12:49,054,880–49,055,046	167	3,896×	✓ Covered
KMT2D	chr12:49,055,256–49,055,344	89	1,769×	✓ Covered
AJUBA				
AJUBA	chr14:22,973,426–22,973,588	163	2,897×	✓ Covered
AJUBA	chr14:22,974,027–22,974,135	109	2,091×	✓ Covered
AJUBA	chr14:22,974,819–22,974,910	92	1,851×	✓ Covered
AJUBA	chr14:22,974,954–22,975,124	171	4,641×	✓ Covered
AJUBA	chr14:22,976,436–22,976,538	103	2,007×	✓ Covered
AJUBA	chr14:22,976,625–22,976,732	108	3,622×	✓ Covered
AJUBA	chr14:22,978,324–22,978,465	142	2,575×	✓ Covered
AJUBA	chr14:22,981,241–22,982,286	1,046	5,683×	✓ Covered
TP53				
TP53	chr17:7,669,592–7,669,710	119	2,258×	✓ Covered
TP53	chr17:7,670,589–7,670,735	147	2,686×	✓ Covered
TP53	chr17:7,673,515–7,673,628	114	2,116×	✓ Covered
TP53	chr17:7,673,681–7,673,857	177	4,702×	✓ Covered
TP53	chr17:7,674,161–7,674,310	150	2,721×	✓ Covered
TP53	chr17:7,674,839–7,674,991	153	2,747×	✓ Covered
TP53	chr17:7,675,033–7,675,256	224	5,229×	● VARIANT DETECTED
TP53	chr17:7,675,974–7,676,292	319	4,317×	✓ Covered
TP53	chr17:7,676,362–7,676,423	62	4,297×	✓ Covered
TP53	chr17:7,676,501–7,676,614	114	3,440×	✓ Covered

FULL PANEL COVERAGE REPORT — Austen, Jane C. (NGS-2026-052104)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
KEAP1				
KEAP1	chr19:10,486,635–10,486,838	204	3,457×	✓ Covered
KEAP1	chr19:10,489,172–10,489,388	217	3,570×	✓ Covered
KEAP1	chr19:10,489,628–10,489,873	246	3,862×	✓ Covered
KEAP1	chr19:10,491,557–10,492,282	726	5,400×	✓ Covered
KEAP1	chr19:10,499,375–10,500,053	679	5,327×	✓ Covered
NFE2L2				
NFE2L2	chr2:177,233,985–177,234,291	307	4,253×	✓ Covered
CASP8				
CASP8	chr2:201,266,467–201,266,811	345	4,484×	✓ Covered
CASP8	chr2:201,271,496–201,271,641	146	2,598×	✓ Covered
CASP8	chr2:201,272,618–201,272,796	179	3,170×	✓ Covered
CASP8	chr2:201,272,878–201,272,962	85	3,889×	✓ Covered
CASP8	chr2:201,274,869–201,274,973	105	1,975×	✓ Covered
CASP8	chr2:201,276,807–201,276,988	182	3,146×	✓ Covered
CASP8	chr2:201,284,796–201,285,337	542	5,126×	✓ Covered
CASP8	chr2:201,286,439–201,286,611	173	3,058×	✓ Covered
CUL3				
CUL3	chr2:224,474,228–224,474,396	169	3,016×	✓ Covered
CUL3	chr2:224,478,180–224,478,365	186	3,236×	✓ Covered
CUL3	chr2:224,481,872–224,482,098	227	3,663×	✓ Covered
CUL3	chr2:224,495,812–224,495,986	175	3,092×	✓ Covered
CUL3	chr2:224,497,733–224,497,869	137	2,545×	✓ Covered
CUL3	chr2:224,500,343–224,500,507	165	2,970×	✓ Covered
CUL3	chr2:224,502,945–224,503,092	148	2,667×	✓ Covered
CUL3	chr2:224,503,632–224,503,842	211	3,498×	✓ Covered
CUL3	chr2:224,505,936–224,506,152	217	3,541×	✓ Covered
CUL3	chr2:224,506,838–224,507,023	186	3,184×	✓ Covered
CUL3	chr2:224,511,334–224,511,602	269	4,121×	✓ Covered
CUL3	chr2:224,513,504–224,513,658	155	2,748×	✓ Covered
CUL3	chr2:224,514,592–224,514,792	201	3,343×	✓ Covered
CUL3	chr2:224,535,508–224,535,661	154	2,725×	✓ Covered
CUL3	chr2:224,557,639–224,557,876	238	3,799×	✓ Covered
CUL3	chr2:224,584,924–224,585,029	106	2,019×	✓ Covered
EP300				
EP300	chr22:41,092,985–41,093,118	134	2,428×	✓ Covered
EP300	chr22:41,117,167–41,117,841	675	5,332×	✓ Covered
EP300	chr22:41,125,844–41,126,060	217	3,618×	✓ Covered
EP300	chr22:41,127,467–41,127,768	302	4,342×	✓ Covered
EP300	chr22:41,129,870–41,130,023	154	2,781×	✓ Covered
EP300	chr22:41,131,368–41,131,653	286	4,155×	✓ Covered
EP300	chr22:41,135,793–41,135,926	134	2,495×	✓ Covered
EP300	chr22:41,137,633–41,137,810	178	3,160×	✓ Covered
EP300	chr22:41,140,120–41,140,277	158	2,830×	✓ Covered
EP300	chr22:41,141,028–41,141,242	215	3,583×	✓ Covered
EP300	chr22:41,146,719–41,146,836	118	2,222×	✓ Covered
EP300	chr22:41,147,817–41,147,966	150	2,721×	✓ Covered
EP300	chr22:41,149,018–41,149,195	178	3,133×	✓ Covered
EP300	chr22:41,149,741–41,150,218	478	4,929×	✓ Covered
EP300	chr22:41,151,813–41,152,032	220	3,625×	✓ Covered
EP300	chr22:41,152,186–41,152,370	185	3,505×	✓ Covered

FULL PANEL COVERAGE REPORT — Austen, Jane C. (NGS-2026-052104)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
EP300	chr22:41,154,975–41,155,133	159	2,853×	✓ Covered
EP300	chr22:41,157,149–41,157,428	280	4,104×	✓ Covered
EP300	chr22:41,158,392–41,158,520	129	2,370×	✓ Covered
EP300	chr22:41,160,622–41,160,742	121	2,279×	✓ Covered
EP300	chr22:41,162,703–41,162,799	97	1,864×	✓ Covered
EP300	chr22:41,164,033–41,164,150	118	2,195×	✓ Covered
EP300	chr22:41,166,579–41,166,686	108	2,099×	✓ Covered
EP300	chr22:41,168,429–41,168,619	191	3,276×	✓ Covered
EP300	chr22:41,168,701–41,168,887	187	4,500×	✓ Covered
EP300	chr22:41,169,483–41,169,636	154	2,756×	✓ Covered
EP300	chr22:41,170,386–41,170,591	206	3,488×	✓ Covered
EP300	chr22:41,172,479–41,172,683	205	3,430×	✓ Covered
EP300	chr22:41,173,603–41,173,804	202	3,427×	✓ Covered
EP300	chr22:41,176,227–41,176,548	322	4,372×	✓ Covered
EP300	chr22:41,176,753–41,178,973	2,221	6,010×	✓ Covered
PIK3CA				
PIK3CA	chr3:179,218,190–179,218,354	165	2,897×	✓ Covered
PIK3CA	chr3:179,234,074–179,234,381	308	4,302×	✓ Covered
FBXW7				
FBXW7	chr4:152,325,986–152,326,251	266	4,034×	✓ Covered
FAT1				
FAT1	chr4:186,588,575–186,589,240	666	5,329×	✓ Covered
FAT1	chr4:186,595,669–186,595,846	178	3,128×	✓ Covered
FAT1	chr4:186,596,520–186,597,191	672	5,307×	✓ Covered
FAT1	chr4:186,597,662–186,597,812	151	2,748×	✓ Covered
FAT1	chr4:186,597,952–186,598,145	194	3,681×	✓ Covered
FAT1	chr4:186,599,878–186,600,380	503	5,045×	✓ Covered
FAT1	chr4:186,601,249–186,601,446	198	3,419×	✓ Covered
FAT1	chr4:186,602,883–186,603,054	172	3,038×	✓ Covered
FAT1	chr4:186,603,156–186,603,997	842	5,680×	✓ Covered
FAT1	chr4:186,604,357–186,604,594	238	3,793×	✓ Covered
FAT1	chr4:186,606,050–186,606,233	184	3,219×	✓ Covered
FAT1	chr4:186,609,163–186,609,340	178	3,132×	✓ Covered
FAT1	chr4:186,609,781–186,610,035	255	3,897×	✓ Covered
FAT1	chr4:186,611,366–186,611,795	430	4,826×	✓ Covered
FAT1	chr4:186,613,089–186,613,362	274	4,097×	✓ Covered
FAT1	chr4:186,614,171–186,614,364	194	3,282×	✓ Covered
FAT1	chr4:186,616,985–186,617,221	237	3,724×	✓ Covered
FAT1	chr4:186,617,688–186,621,795	4,108	6,161×	✓ Covered
FAT1	chr4:186,628,134–186,628,384	251	3,960×	✓ Covered
FAT1	chr4:186,628,468–186,628,783	316	5,087×	✓ Covered
FAT1	chr4:186,633,664–186,633,843	180	3,210×	✓ Covered
FAT1	chr4:186,636,005–186,636,255	251	3,927×	✓ Covered
FAT1	chr4:186,636,565–186,636,934	370	4,618×	✓ Covered
FAT1	chr4:186,639,702–186,639,803	102	1,944×	✓ Covered
FAT1	chr4:186,663,279–186,663,633	355	4,524×	✓ Covered
FAT1	chr4:186,706,543–186,709,847	3,305	6,130×	✓ Covered
TERT				
TERT	chr5:1,295,050–1,295,250	201	3,370×	✓ Covered
PIK3R1				
PIK3R1	chr5:68,294,516–68,294,698	183	3,145×	✓ Covered

FULL PANEL COVERAGE REPORT — Austen, Jane C. (NGS-2026-052104)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
PIK3R1	chr5:68,295,128-68,295,344	217	3,611x	✓ Covered
PIK3R1	chr5:68,295,400-68,295,508	109	4,581x	✓ Covered
NSD1				
NSD1	chr5:177,135,084-177,136,050	967	5,635x	✓ Covered
NSD1	chr5:177,191,864-177,192,039	176	3,103x	✓ Covered
NSD1	chr5:177,204,100-177,204,312	213	3,556x	✓ Covered
NSD1	chr5:177,209,616-177,212,215	2,600	6,066x	✓ Covered
NSD1	chr5:177,235,801-177,235,965	165	2,940x	✓ Covered
NSD1	chr5:177,238,217-177,238,527	311	4,327x	✓ Covered
NSD1	chr5:177,239,736-177,239,885	150	2,709x	✓ Covered
NSD1	chr5:177,244,175-177,244,290	116	2,172x	✓ Covered
NSD1	chr5:177,246,658-177,246,816	159	2,848x	✓ Covered
NSD1	chr5:177,248,161-177,248,344	184	3,214x	✓ Covered
NSD1	chr5:177,251,710-177,251,873	164	2,883x	✓ Covered
NSD1	chr5:177,256,931-177,257,171	241	3,818x	✓ Covered
NSD1	chr5:177,259,969-177,260,188	220	3,628x	✓ Covered
NSD1	chr5:177,267,542-177,267,738	197	3,397x	✓ Covered
NSD1	chr5:177,269,582-177,269,827	246	3,871x	✓ Covered
NSD1	chr5:177,273,652-177,273,804	153	2,727x	✓ Covered
NSD1	chr5:177,280,545-177,280,854	310	4,305x	✓ Covered
NSD1	chr5:177,282,445-177,282,601	157	2,823x	✓ Covered
NSD1	chr5:177,283,767-177,283,948	182	3,179x	✓ Covered
NSD1	chr5:177,288,799-177,288,945	147	2,639x	✓ Covered
NSD1	chr5:177,291,934-177,292,178	245	3,828x	✓ Covered
NSD1	chr5:177,293,812-177,295,476	1,665	5,907x	✓ Covered
EGFR				
EGFR	chr7:55,173,901-55,174,063	163	2,968x	✓ Covered
EGFR	chr7:55,174,702-55,174,840	139	2,528x	✓ Covered
EGFR	chr7:55,181,273-55,181,498	226	3,647x	✓ Covered
EGFR	chr7:55,191,699-55,191,894	196	3,351x	● VARIANT DETECTED
CDKN2A				
CDKN2A	chr9:21,968,212-21,968,262	51	1,121x	✓ Covered
CDKN2A	chr9:21,970,882-21,971,228	347	4,453x	✓ Covered
CDKN2A	chr9:21,974,658-21,974,846	189	6,561x	✓ Covered
CDKN2A	chr9:21,974,658-21,974,847	190	6,579x	✓ Covered
NOTCH1				
NOTCH1	chr9:136,496,054-136,497,578	1,525	5,881x	✓ Covered
NOTCH1	chr9:136,498,879-136,499,016	138	2,545x	✓ Covered
NOTCH1	chr9:136,499,092-136,499,279	188	4,444x	✓ Covered
NOTCH1	chr9:136,500,532-136,500,867	336	4,352x	✓ Covered
NOTCH1	chr9:136,501,728-136,501,933	206	3,430x	✓ Covered
NOTCH1	chr9:136,501,981-136,502,108	128	4,865x	✓ Covered
NOTCH1	chr9:136,502,252-136,502,508	257	4,243x	✓ Covered
NOTCH1	chr9:136,503,162-136,503,350	189	3,307x	✓ Covered
NOTCH1	chr9:136,504,653-136,505,124	472	4,942x	✓ Covered
NOTCH1	chr9:136,505,290-136,505,901	612	5,282x	✓ Covered
NOTCH1	chr9:136,506,507-136,506,659	153	2,820x	✓ Covered
NOTCH1	chr9:136,506,696-136,506,993	298	5,557x	✓ Covered
NOTCH1	chr9:136,507,285-136,507,457	173	3,040x	✓ Covered
NOTCH1	chr9:136,507,935-136,508,159	225	3,651x	✓ Covered
NOTCH1	chr9:136,508,212-136,508,405	194	5,090x	✓ Covered

FULL PANEL COVERAGE REPORT — Austen, Jane C. (NGS-2026-052104)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
NOTCH1	chr9:136,508,850-136,509,091	242	3,819×	✓ Covered
NOTCH1	chr9:136,509,713-136,509,981	269	4,010×	✓ Covered
NOTCH1	chr9:136,510,633-136,510,825	193	3,374×	✓ Covered
NOTCH1	chr9:136,511,132-136,511,291	160	2,872×	✓ Covered
NOTCH1	chr9:136,513,001-136,513,154	154	2,779×	✓ Covered
NOTCH1	chr9:136,513,372-136,513,557	186	3,189×	✓ Covered
NOTCH1	chr9:136,514,490-136,514,722	233	3,742×	✓ Covered
NOTCH1	chr9:136,515,270-136,515,420	151	2,768×	✓ Covered
NOTCH1	chr9:136,515,463-136,515,736	274	5,327×	✓ Covered
NOTCH1	chr9:136,515,961-136,516,114	154	2,799×	✓ Covered
NOTCH1	chr9:136,517,252-136,517,405	154	2,770×	✓ Covered
NOTCH1	chr9:136,517,732-136,517,957	226	3,716×	✓ Covered
NOTCH1	chr9:136,518,117-136,518,312	196	3,637×	✓ Covered
NOTCH1	chr9:136,518,571-136,518,844	274	4,058×	✓ Covered
NOTCH1	chr9:136,519,423-136,519,585	163	2,892×	✓ Covered
NOTCH1	chr9:136,522,830-136,523,208	379	4,569×	✓ Covered
NOTCH1	chr9:136,523,697-136,523,999	303	4,258×	✓ Covered
NOTCH1	chr9:136,544,004-136,544,122	119	2,258×	✓ Covered
NOTCH1	chr9:136,545,706-136,545,806	101	1,986×	✓ Covered